JASMINE: Jordanian Autism Screening using Multimodal Intelligent Neurodevelopmental Evaluation

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Abstract

The JASMINE System is a web-based intelligent autism screening and assessment platform. The goal of the system is to assist in early detection of Autism Spectrum Disorder (ASD) in children.

The system was created to be used by rehabilitation specialists and therapists working in early childhood developmental assessment. The system was created to be available on a secure web-based platform which makes the system accessible at all times and anywhere by simply using a web browser and Internet connectivity.

This research project presents the main steps that led to the creation of the system, starting from gathering data about different reporting systems that were available on the Internet and reviewing them. Then, specifying the requirements using Use Case analysis method that was needed to accomplish the design and development phase of the system. The last step was implementing the system.

Acknowledgments

We would like to extend our heartfelt gratitude to the individuals who have made notable contributions to this research. We convey our sincere thanks to our supervisor Dr. Amera Al-Amery for her invaluable effort in guiding us throughout this research. Her profound knowledge and unwavering commitment greatly enhanced the breadth and excellence of our work. In addition, we express our gratitude to the wider team, friends, family, research community, and the open-source community for their consistent and steadfast support during this undertaking. Last but not least, we would like to express our pleasure to be part of Jordan University of Science and Technology. We deeply appreciate their indispensable effort of providing the necessary platforms and resources that significantly influenced the results of our project.

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1. Chapter 1
Introduction

1.1. Introduction

1.1.1. AI in the Medical Field

The rise of artificial intelligence in healthcare has been revolutionary, completely reshaping how patients are diagnosed, treated, and monitored. The integration of AI in the field has led to a transformation in the medical field, one that is reflected in the improved accuracy of disease diagnosis and the impressive drop in response time. This is all possible due to AI systems' ability to process large volumes of data with remarkable accuracy, helping physicians detect patterns that may not be easily visible through human observation alone. The potential of AI in healthcare is nothing short of remarkable. Experts predict that artificial intelligence in healthcare will continue to redefine how we process clinical data, diagnose complex conditions, develop breakthrough treatments, and even prevent diseases before they occur.

1.1.2. AI in Aiding Early Diagnosis

In many critical disease cases, treatment is much more effective if the disease is detected early. In an ideal world, symptoms would present themselves early enough and allow for ample time to seek help. However, that is not the case and often we may not know we are sick until the disease is advanced. That is not to say that diseases don't form gradually; the symptoms that emerge in the early stages are frequently subtle, nonspecific, or easily mistaken for minor, everyday discomforts, and thus, can easily be overlooked when studied by a human. As a result of such challenges, artificial intelligence has emerged as a powerful tool in this regard.

1.1.3. Autism Spectrum Disorder (ASD)

Neurodevelopmental disorders affect thousands of millions of children worldwide. Autism, scientifically known as autism spectrum disorder (ASD), is a neurodevelopmental condition characterized by a wide range of challenges in social communication, language, behavior, and social interaction. The manifestations of autism vary widely, giving rise to the concept of the "spectrum", which includes individuals with mild to severe symptoms. Signs of autism can emerge from early childhood but are often identified in preschool or school age, when they become more evident. ASD comes with its fair share of indicators, most of which go unnoticed by parents and specialists, due to their subtle and tenuous nature. Usually, diagnosis requires extensive observation by trained specialists.

1.1.4. Autism In Jordan

Autism spectrum disorder (ASD) in Jordan presents significant challenges: public awareness remains limited, with a recent survey reporting an average knowledge score of only 36.5% among Jordanians. Diagnostic and rehabilitation services are constrained by cost, transportation barriers, and a shortage of specialized centers. Screening is currently manual and largely dependent on parent-report questionnaires and clinician-administered scales rather than automated or digital tools. Moreover, diagnostic practices lack uniformity: different centers use diverse assessment methods and forms, and there is no national standard pathway for diagnosis. In terms of locally validated instruments, Jordan has adapted the **Autism Spectrum Rating Scale (ASRS-3)** for children aged 6 to 18, which demonstrates strong psychometric properties in the Jordanian context. For toddlers, a Jordanian version of the **Modified Checklist for Autism in Toddlers–Revised (M-CHAT-R/F)** has also been validated, showing good sensitivity and specificity.

1.1.5. AI & Autism

As with all diseases, timely diagnosis is crucial for early intervention, allowing the afflicted children a chance to reduce challenging behaviors and improve learning skills through proper support and rehabilitation. In recent years, artificial intelligence has found its way into the field of autism screening, which has opened new possibilities for identifying developmental differences much earlier and with greater accuracy.

Artificial intelligence has begun to augment traditional autism screening by providing objective, scalable methods to detect early markers of ASD from behavioral and digital data. Early work demonstrated that automated analysis of home videos and mobile-recorded gaze patterns can distinguish children at elevated risk for ASD, suggesting that computer vision and machine-learning approaches can capture subtle social attention and affective cues missed by routine observation. More recent reviews and studies have extended these findings to automated pipelines that combine facial expression, gaze, gesture, and acoustic features with classical parent-report instruments, reporting promising classification performance across clinical and community samples.

These advances indicate that AI-enabled screening can complement, rather than replace, clinical assessment by prioritizing children for formal diagnostic evaluation and by enabling more equitable, scalable identification pathways.

1.2. Research Motivation

With the advance of artificial intelligence, mainly computer vision, we see the potential in utilizing such technology in assisting healthcare professionals detect ASD indicators efficiently and objectively. This project will delve into the use of computer vision to analyze children's body posture, movement patterns, and behavioral cues, with the purpose of helping specialists in the initial screening process.

1.3. Background

Traditionally, screening for autism and other neurodevelopmental conditions relies heavily on extensive monitoring done by specialists. Another drawback is that these assessments take several

hours and children may have to undergo these assessments at different ages for accurate diagnosis. While effective, these methods are time-consuming, stressful, and prone to human bias.

Propelled with the rise in use of machine learning techniques in the research dimensions of medical diagnosis, new opportunities of early diagnosis remain unexplored. By leveraging such techniques, it is possible to extract meaningful patterns from visual data that may correlate with behavioral markers of neurodivergence.

1.4. Problem Statement

In Jordan, more generally the MENA region, awareness and understanding of autism and neurodivergence are still developing, and social stigma often discourages families from seeking professional diagnosis or intervention. As a result, many children may remain undiagnosed or receive late diagnoses, limiting their access to early support programs.

The current dependence on manual evaluation limits the ability of specialists to screen large numbers of children effectively. Furthermore, the limited number of specialists in some regions creates additional barriers to early detection. This project aims to address these challenges by developing an AI-assisted screening tool that analyzes behavioral and postural patterns to support specialists in identifying early signs of neurodivergence, ultimately reducing the gap between observation and diagnosis.

1.5. Importance of the project

The importance of this project lies in its contribution to a field that remains a mystery to the millions of people affected by it. It aims to improve the early detection and support of neurodivergent children, while also addressing the challenges faced by families and specialists during the diagnostic process. In Jordan, parents and specialists often experience frustration due to the lengthy, costly, and emotionally exhausting procedures required to obtain an accurate diagnosis. Children may undergo multiple assessments over extended periods, delaying access to early intervention and support services.

By utilizing artificial intelligence to analyze posture and behavioral cues, this project aims to offer a more efficient and accessible preliminary screening method. The system provides data-driven insights that can assist specialists in identifying potential developmental differences more quickly and objectively. Early and accurate screening not only reduces the burden on families and professionals but also enables timely intervention, which has been shown to significantly improve long-term outcomes for children on the autism spectrum.

Furthermore, the project fosters collaboration between the fields of computer science and healthcare, demonstrating how technology can enhance the quality, efficiency, and accessibility of neurodevelopmental assessment. This project is important because it contributes to both the technological and social aspects of autism awareness and diagnosis in Jordan. From a technical perspective, it applies data science and AI to an area of significant public health impact, demonstrating how modern technologies can enhance the work of rehabilitation specialists. From a social standpoint, it encourages early and objective screening, helping reduce stigma by framing diagnosis as a data-informed and supportive process.

The collaboration between academic research and rehabilitation centers promotes a stronger link between technology and community health services, paving the way for future innovations in inclusive healthcare.

1.6. Required Tools

Table 1: Required Tools

Tool Name	*Description*
Python	Python was selected as the primary development language for JASMINEproject due to its extensive ecosystem for data science and machine learning. As highlighted in the literature review, the majority of modern ASD screening research relies on Deep Learning (DL) and Computer Vision (CV), domains where Python is the industry standard
R	R is intended to be used for the post-processing and statistical validation of the system's results. Specifically, R will be utilized to analyze the "Risk Scores" and "Confidence Levels" generated by the AI model. It will facilitate the calculation of performance metrics such as the required "minimum sensitivity/recall of 80%" and help determine the statistical significance of the model's predictions compared to standard clinical baselines
OpenCV	OpenCV (Open Source Computer Vision Library) is a massive open-source library designed for real-time computer vision and image processing tasks, It is utilized for Video Preprocessing. Before analysis, OpenCV handles the reading of video files, frame extraction, resizing, and color conversion to prepare the raw footage for the AI models.
StreamLit	To meet the stakeholder requirement for a "simple, web-based interface that does not require coding knowledge", Streamlit was chosen as the frontend framework. It allows for the rapid deployment of the "Dashboard" and facilitates the secure "Drag and Drop" of video files, ensuring the system is accessible to non-technical rehabilitation specialists.
OpenPose	OpenPose is a real-time multi-person system to jointly detect human body, hand, facial, and foot keypoints (in total 135 keypoints) on single images. OpenPose serves as a robust alternative for the Pose and Motion Extraction module. It is capable of analyzing the uploaded video footage to extract detailed skeletal data, including the precise position of the child's limbs and head orientation, which are critical inputs for the AI model to detect repetitive behaviors or posture abnormalities. It was considered for the project due to its high accuracy in detecting keypoints, even in challenging scenarios with complex backgrounds or when multiple people (e.g., a parent and child) are present in the frame.
Overleaf	Overleaf is a collaborative scientific writing platform commonly used in multidisciplinary research projects. It supports version control and consistent academic formatting. In this project, Overleaf facilitates structured documentation and collaboration between team members, ensuring alignment between system design, implementation decisions, and research reporting.
MediaPip	To fulfill the non-functional requirement of analyzing videos without physical sensors, we utilized MediaPipe. This framework allows for robust skeletal keypoint

	extraction, enabling the system to track the child's movement and posture from standard RGB video footage
TensorFlow	TensorFlow serves as the foundational computational engine for the AI Behavior Analysis Module. It handles the heavy lifting of tensor operations and mathematical computations required to train the Deep Learning models that process the video data and calculates the ASD "Risk Score". TensorFlow efficiently manages the complex matrix multiplications required by the Convolutional Neural Networks (CNNs) and LSTM networks identified in the literature review as effective for autism screening

2. Chapter 2

Literature review

2.1. Research Articles

Table 2: Related Research Matrix

ID	*Research Paper*	*Data*	*Data size*	*Model(s) Used*	*Evaluation Method(s)*	*Best Result*	*Result*
R1	Analysis and Detection of Autism Spectrum Disorder Using Machine Learning Techniques	UCI Repository First, (children): 292 records, 21 features Second, (adult): 705 records, 21 features Third, (adolescent): 104 records, 21 features	Total Of: 1100 records and 63 features	1. ML 2. SVM 3. NB 4. LR 5. CNN 6. ANN	Specificity, Sensitivity and accuracy	CNN accuracy: 99.53% adults 98.30% children 96.88% adolescents	"These results strongly suggest that a CNN based model can be implemented for detection of Autism Spectrum Disorder instead of the other conventional machine learning classifier suggested in earlier research."
R2	Autism Spectrum Disorder Screening on Home Videos Using Deep Learning	Videos collected from Youtube	unknown	1. CNN 2. MobileNet 3. LSTM, 4. Bi-LSTM	accuracy, precision, recall and F1-score.	MobileNet + Bi-LSTM Accuracy: 84.95% precision, recall, and F1-score: 89%	The experiment results on various combinations of pre-trained CNN models and LSTM/BiLSTM show that the proposed model can be used as an initial autism screening tool. Among the different

							combinations, the MobileNet and Bi-LSTM combo achieved the best test accuracy of 84.95% with 89% precision, recall and F1-score.
R3	Detecting Developmental Delay and Autism Through Machine Learning Models Using Home Videos of Bangladeshi Children: Development and Validation Study	159 videos collected via mobile phones. 150 children, ages 18 months-4 years, 2-5 minute videos	159 videos	1. Logistic Regression with Elastic Net Regularization 2. Random Forest 3. SVM	Accuracy, Sensitivity, Specificity, Area Under the Curve (AUC), Unweighted Average Recall	Layer 1 of the classifier distinguished typically developing children from those with atypical development with 70% accuracy and an AUC of 76%, while Layer 2 outperformed it in differentiating ASD from other conditions by achieving 76% accuracy and an AUC of 85% with balanced sensitivity and specificity."	Achieved an accuracy (area under the curve [AUC]) of 76% (SD 3%) and sensitivity of 76% (SD 4%) for identifying atypical children from among developmentally delayed children, and an accuracy (AUC) of 85% (SD 5%) and sensitivity of 76% (SD 6%) for identifying children with ASD from those predicted to have other developmental delays.
R4	Mobile detection of autism through machine learning on homevideo: A development and prospective validation	162, 116 ASD, 46 TD	162 videos	1. ADTree8 2. ADTree7 3. SVM5 4. LR5 5. LR9 6. SVM12 7. SVM10 8. LR10	accuracy,AUC	"The top-performing classifier was LR5, which showed an accuracy of 88.9%, sensitivity of 94.5%, and specificity of 77.4%."	"LR5 exhibited high accuracy on all age ranges with the exception of children over 6 years old (although note that we had limited examples of non-ASD [n = 1] class in this range). This model performed best on children between the ages of 4 and 6 years, with sensitivity and specificity both above 90%"

	study						
R5	Early Detection of Autism Spectrum Disorder Through Video Based Movement and Posture Analysis using Deep Learning	GIF, videos, pose estimation data, audio, and depth information	1,315 data samples	1. CNN 2. LSTM 3. Bi-LSTM 4. Attention Mechanism 5. MediaPipe Pose Estimation	Accuracy, AUC-ROC, Cohen's	Hybrid CNN + Bi-LSTM + Attention: ~86.4% ± 1.3% (5-fold cross validation)	Hybrid CNN-LSTM with Attention + Mediapipe pose Estimator, outperforming 3D-ResNet models and other referenced baselines.Proposed model is 3.2× faster 12.7% accuracy improvement over baseline 3D-CNN
R6	EXPLAINABLE CNN-BASED ADHD DETECTION USING EEG DATA	Electroencephalography (EEG)	unknown	1. CNN 2. LIME 3. SHAP	Accuracy	CNN accuracy: 98.91%	"..., the model demonstrates robust performance, achieving an accuracy of 98.91% on unseen test data. "
R7	Machine Learning Based Autism Spectrum Disorder Detection from Videos	infants aged 6-36 months, directed gaze towards faces. 1707 videos of 3 min duration	1707 videos	ResNet-18, two independent neural networks	Accuracy	Results of facial key points based method: (Look face task) sensitivity 0.59 specificity 0.73 (smile detection) specificity 0.92 sensitivity 0.45 Results of image based method: F1-score,	"baseline ML classifier to achieve an accuracy of 82% for ASD diagnosis." "We achieve 70% accuracy for smile, 68% accuracy for look face, 67% for look object and 53% accuracy for vocalization."

						macro-average both greater than 67%	
R8	One Size Does Not Fit All: Considerations when using Webcam-Based Eye Tracking to Models of Neurodiverget Learners' Attention and Comprehensio n	webcam-based eye tracking	354	WebGazer(data collection tool)	Kappa, and Area Under the Receiver Operating Characteris tic Curve (AUROC)	Neurotypical, count=171, Feature combination=NLP, Model=XGB, Base rate=0.79, Kappa=0.29, AUROC=0.71	"Results indicate that diagnosis-specific models provide more accurate predictions (AUROC's .59-.70 vs. .57 for the general model),"
R9	The Incorporation of Artificial Intelligence in the Identification of Neurological Disorders	MRI scan	unknown	1. CNN 2. MLP 3. RNN	Kappa, and Area Under the Receiver Operating Characteris tic Curve (AUROC)	Accuracy	"..., it was hypothesized that the CNN model would be the most efficient. ..., it was found that [CNN] had the quickest response time and the most accuracy compared to the other models."

R10	Video-audio neural network ensemble for comprehensive screening of autism spectrum disorder in young children	160 children ages 1-5	160 video and audio	"...a multi-modal approach by combining two neural networks trained on video and audio features..."	accuracy, F1 score, Precision, Recall, sensitivity, specificity.	specificity (92.5%, i.e., few false negatives) or sensitivity (90%, i.e. few false positives)."	"...an accuracy of 82.5% (F1 score: 0.816, Precision: 0.775, Recall: 0.861) for screening Autism Spectrum Disorders (ASD)." "Additional combinations of our model were developed to achieve higher specificity (92.5%, i.e., few false negatives) or sensitivity (90%, i.e. few false positives)."
R11	Video-Based Autism Detection with Deep Learning	video data, the participants sit down and look at a screen, towards the camera, with a neutral face. The stimulus item is then provided, the participant interacts with it, and then the camera captures the reactions–the movements of the subject, their head, and their facial expressions.	There are 150 taste experiment videos available to us, totaling approximately 98,000 frames, or 653 frames per video, on average. There are 240 smell experiment videos available, totaling approximately 153,000 frames, or 637 frames per video	DL, two convolutional neural network (CNN), temporal transformer, Pytorch, AdamW optimizer	F1 score, accuracy.	EfficientNet-B0 + ResNet-18 + Temporal Transformer Accuracy: 81.48% F1-score: 0.7289	"In our experiments, we sample 16 frames twice, thus only 32 frames per video. Sampling two times gives a good trade-off between speed, memory consumption, and performance. " "The test accuracy on the test dataset is 81.48% with an F1 score of 0.7289. The result indicates the model is able to generalize well to similar but unseen samples, despite limited data and the number of frames processed per video."

R12	Using Machine Learning for Motion Analysis to Early Detect Autism Spectrum Disorder: A Systematic Review	body motion, eye/gaze behaviour, inertial sensor data.	unknown	this article is a review it does not have any models	accuracy	SVM model: accuracy=96.7%	"The literature review showed that motion patterns ML analysis predicts ASD classification as accurately as classical gold-standard tools."
R13	Early Diagnosis of Autism: A Review of Video-Based Motion Analysis and Deep Learning Techniques	Publicly Available ASD Behavioral Datasets (videos, PAVIS dataset, MMBD (Multimodal) dataset,...etc)	2447 videos	VGG16, ConvNeXt-T, and ResNet-50. ResNet-50	classificatio n accuracy	Using Support Vector Machine (SVM) on kinematic data: classification accuracy up to ~96.7%.	"We categorize the leading research according to the body parts involved: head pose, hand and arm pose, and trunk pose."
R14	MMASD: A Multimodal Dataset for Autism Intervention Analysis	32 children with ASD; 1,315 data samples; 11 activity classes	32 children with ASD; 1,315 data samples	N/A (Dataset paper)	N/A (Dataset paper)	N/A	Introduces a publicly accessible, privacy-preserving multimodal dataset for ASD intervention analysis

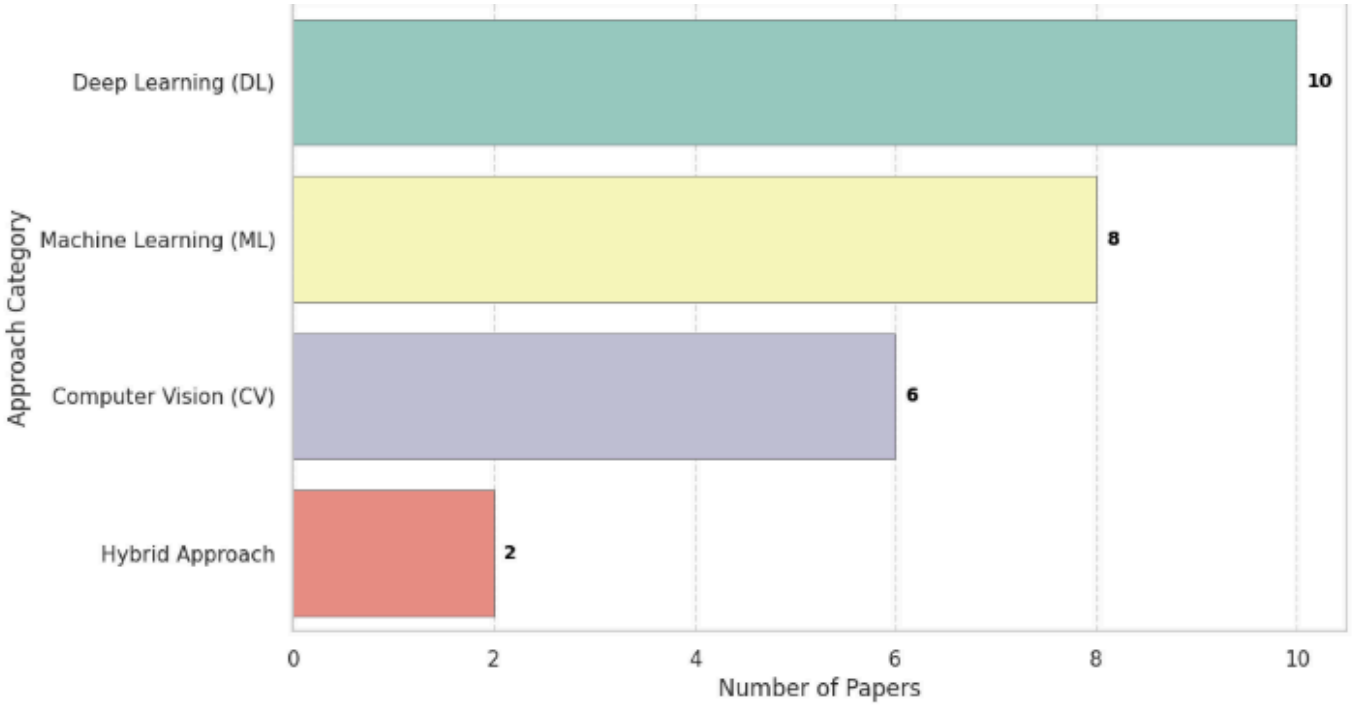
R15	The Potential of Accelerating Early Detection of Autism through Content Analysis of YouTube Videos	100 public YouTube videos children ages 1-15 ASD=45 NON-ASD=55	100 videos	NO MODELS USED This research sees whether it is possible to apply ADOS1 using video analysis on YT videos using human raters. it does not use any computational models any models	Classificatio n, accuracy, sensitivity, specificity	Classification accuracy 96.8% sensitivity 94.1% specificity 100%	" The high classification accuracy (96.8%) supported the feasibility of applying ADOS module 1 questions to videos on YouTube." " Our results show high classification accuracy and inter-rater reliability and together demonstrate that the ADOS module 1 questions can be used by non-clinicians on unstructured videos to effectively distinguish behavioral differences among children with and without ASD"
R16	Developing a New Autism Diagnosis Process Based on a Hybrid Deep Learning Architecture Through Analyzing Home Videos	the AGRE/ADOS database, Kaggle database, and a self-gathered video test dataset with corresponding ADOS data	unknown	Logistic regression, SVM, CNN, DenseNet	accuracy	accuracy 84%	"Our proposed hybrid architecture in this research is able to effectively speed up the diagnosis of ASD without compromising clinical performance. Our study is able to diagnose ASD at an effective accuracy, 84%, with balanced sensitivity and precision, while combining results from 2 separate models"
R17	MMASD+: A Novel Dataset for Privacy-Preserving Behavior Analysis of Children with Autism Spectrum	3D Skeleton coordinates, 3D Body Mesh (ROMP videos), Optical Flow; Derived from raw therapy videos	https://github.com/pavanravva/Enhanced-MMASD	Multimodal Transformer (ViViT for ROMP videos, 3D-CNN for Optical Flow, LSTM for skeleton coordinates, Multi-head	accuracy	96.8% accuracy, 0.97 F1 for action classification, 96.42% accuracy and 0.9362 F1 for ASD identification	Enhanced dataset addresses original MMASD limitations, Proposed framework achieves high accuracy in action classification and ASD identification, outperforms data with single modality

	Disorder			attention fusion)			
R18	Use of Artificial Intelligence to Shorten the Behavioral Diagnosis of Autism	Autism Diagnostic Interview–Revised (ADI-R). (structured caregiver interviews)	894 individuals, between autistic individuals and non autistic.	Decision trees, Rule-based classifiers, Nearest-neighbor–based methods	Accuracy	ADTree accuracy: 99.90%.	"In both cases [SSC and AC validation], our classifier performed with nearly 100% statistical accuracy, properly categorizing all but one of the individuals … who previously had been diagnosed with autism through the standard ADI-R." "Our analysis showed that 7 of the 93 items contained in the ADI-R were sufficient to classify autism with 99.9 % statistical accuracy."
R19	Multi-modular AI Approach to Streamline Autism Diagnosis in Young Children	Parent-report questionnaire collected from a mobile app, key behaviors monitored from home videos (semi-structured data), clinician questionnaire	n=375	Custom ML models, Probabilistic combination of three modules	AUC, Accuracy, Sensitivity	AUC 0.92	Multi-modular assessment outperforms baselines (M-CHAT-R, SRS, CBCL) by 0.35 AUC and 0.69 specificity at 90% sensitivity overall; For <48 months: 0.18 AUC and 0.30 specificity improvement; Allows inconclusive outcomes for higher accuracy (AUC up to 0.92)

R20	AI-assisted early screening, diagnosis, and intervention for autism in young children	data from other studies(videos, eye tracking,...etc)	unknown	CNN, AI-based Diagnostic Device, alternating decision tree, Deep Learning Models.	unknown	Alternating Decision Tree model (behavior-based): 99.97% accuracy for diagnosing autistic children aged 13–48 months.	“the output showed … The model effectively diagnosed autistic children aged 13 to 48 months with an accuracy of 99.97%.” “an autism screening system … worked well for young children … with an accuracy of 97.95%, a sensitivity of 95.53% and a specificity of 98.63%.”
R21	Automated AI based identification of autism spectrum disorder from home videos	510 children (253 ASD, 257 typically developing), aged 18–48 months	Videos from 510 children	LightGBM, Logistic Regression	AUC-ROC, accuracy	AUC-ROC = 0.83 accuracy = 0.75	"The ensemble model, integrating predictions across videos from multiple tasks, achieved an AUROC of 0.80 at baseline, increasing to 0.83 with metadata inclusion. This ensemble approach provided the most robust and generalizable classification performance, underscoring the benefit of aggregating diverse behavioral dimensions. External validation on noisy video samples achieved an AUROC of 0.73, supporting the feasibility of applying the model under variable home-recording conditions. "

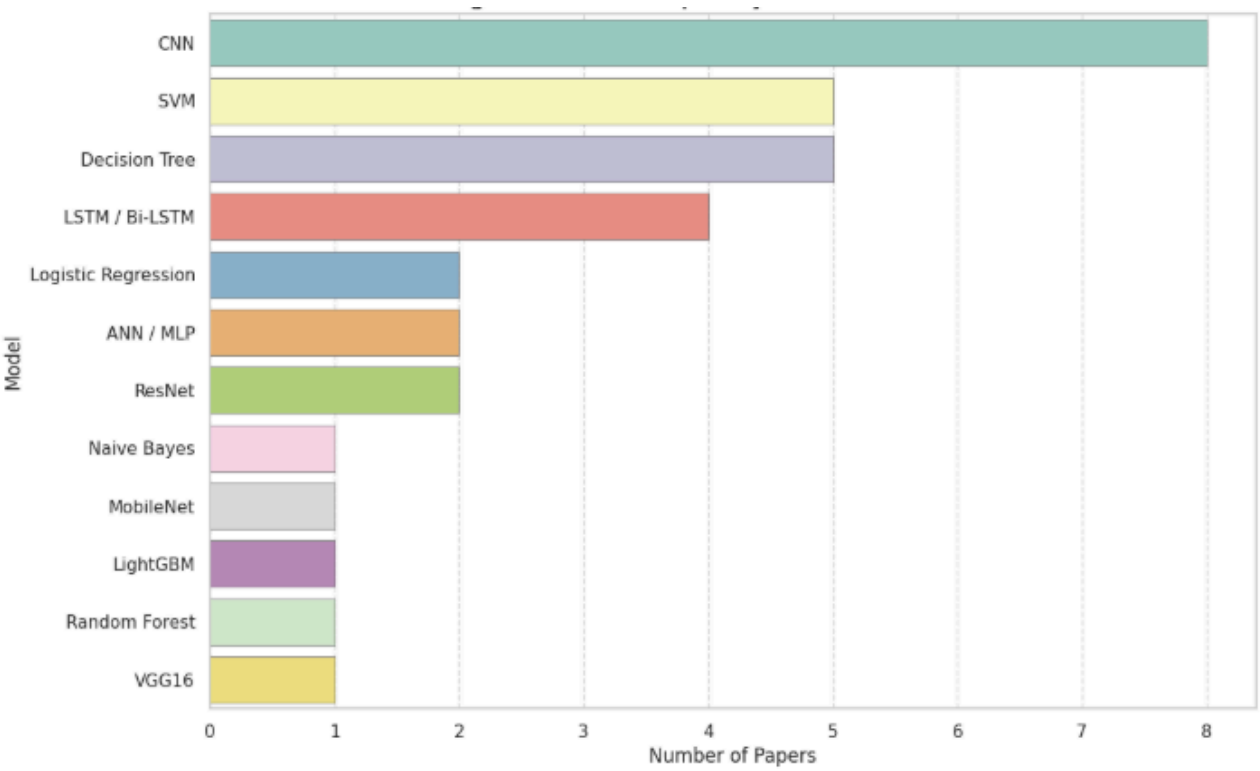
2.2. Literature Discussion

Figure 1: Research Paper Approaches



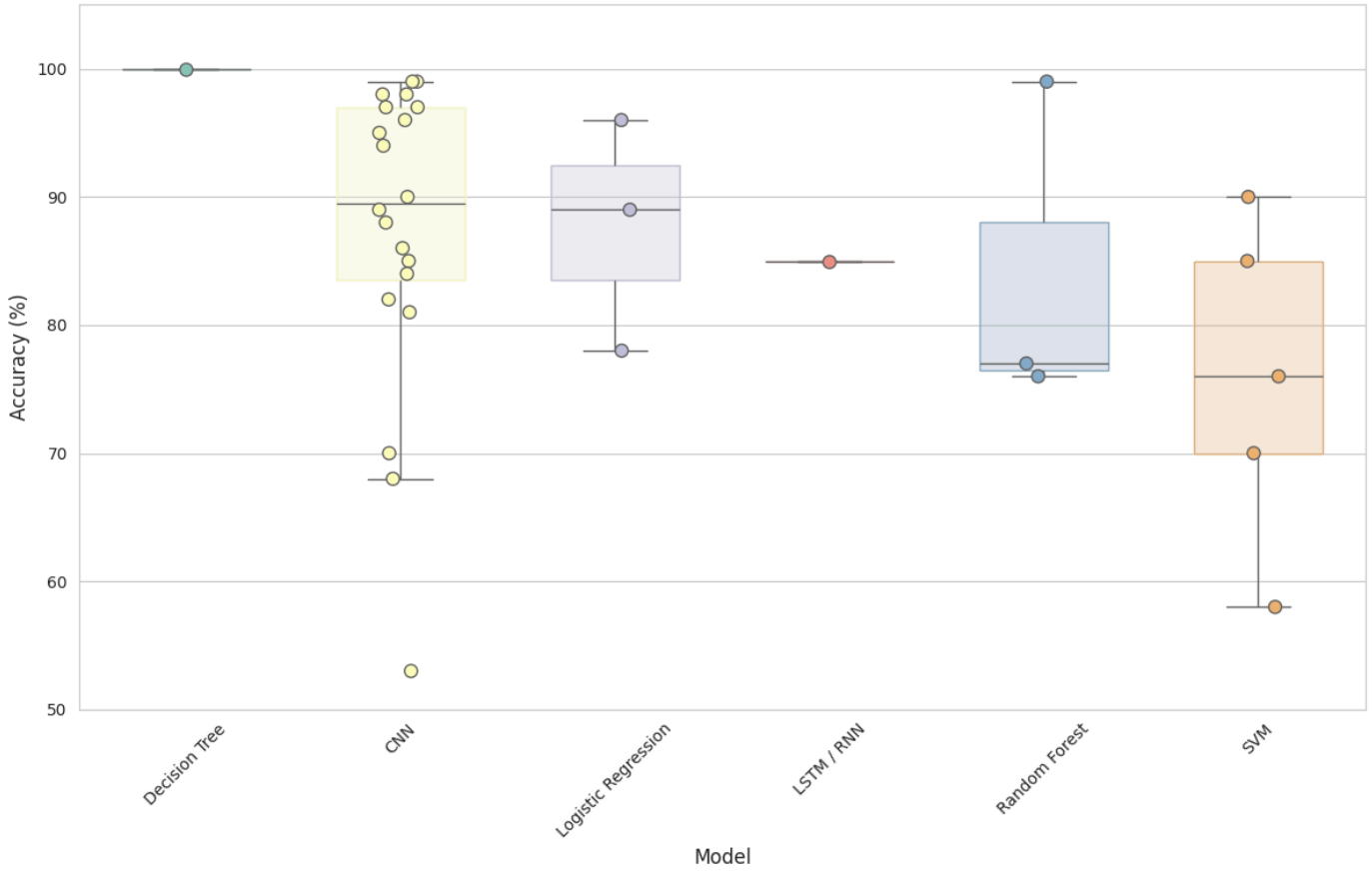
Deep Learning (DL) The most dominant strategy (10 papers), reflecting the modern trend towards using CNNs and LSTMs for complex ASD detection, Machine Learning (ML) Remains a strong competitor (8 papers), particularly for tabular data where models like SVM and Decision Trees, Computer Vision (CV) Explicitly mentioned in 6 papers, highlighting the focus on visual behavioral analysis (eye tracking, movement), Hybrid Approach: A smaller but significant subset (2 papers) combining methods for robust results.

Figure 2: Models used in papers in Literature Review



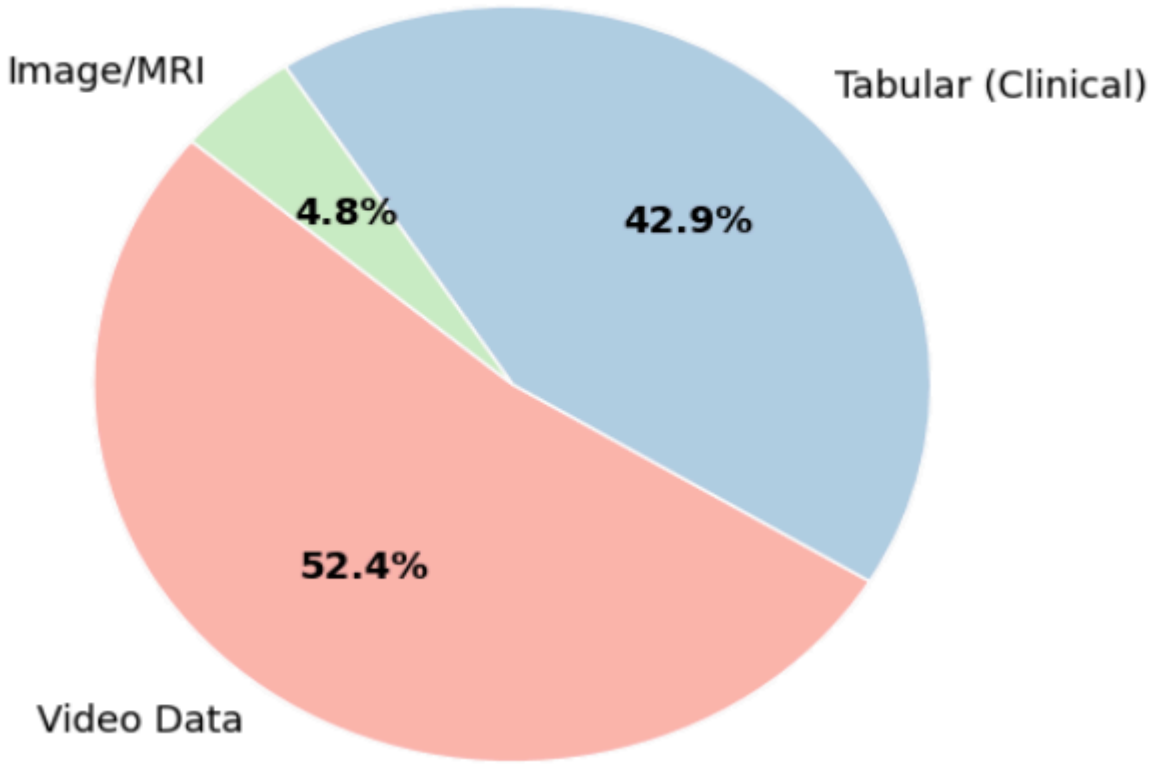
From the chart it shows that 8 papers used the CNN (Convolutional Neural Network) model in their research paper, 5 papers used the SVM (supervised machine learning) model in their research paper, 5 papers used Decision Tree, 4 papers usedLSTM/Bi-LSTM (Long Short-Term Memory/Bidirectional LSTM), 3 papers used Logistic Regression, 3 papers used ANN/MLP (Artificial Neural Network/Multilayer Perceptron), 2 papers used ResNet (Residual Network), 1 paper used Naive bayes model, 1 paper used MobileNet model, 1 paper used LightGBM model, 1 paper used Random Forest model and 1 paper used VGG16 model in their paper.

Figure 3: Average Accuracy of Models used in the papers



The analysis reveals a trade-off between model stability and peak performance. While simpler models like Decision Trees demonstrate nearly perfect consistency (99.9%) on structured clinical tests, sophisticated Deep Learning models (CNN) exhibit significant variability (ranging from 53% to 99%) alongside their high median accuracy (~87%), reflecting the challenge of analyzing complex, unstructured behavioral data (video/images). Despite this variance, they remain critical for automated, non-invasive ASD screening systems. Logistic Regression maintained a strong median accuracy of ~89%, while Random Forest displayed a wider spread with a median of ~77%, and SVM indicated a median accuracy of ~76% with notable performance fluctuations across the studies reviewed.

Figure 4: Data Types in research papers



It shows that up to 42.9% of the data used in the papers are Tabular (Clinical), while up to 52.4% are data taken from videos and then up to 4.8% of the data from the papers are taken from Image/MRI, The distribution suggests that while traditional clinical records remain fundamental, **video-based data** has emerged as a dominant source of information, even surpassing structured clinical data.

2.3. Tools Background

Table 3: Tools Background

Tool Category / Technology	*Background Usage in ASD Screening Research*
Feature Extraction Technologies	Feature extraction technologies are widely used in autism screening research to convert raw data, such as videos or motion sequences, into meaningful numerical representations. In ASD-related studies, extracted features may include joint angles, movement velocity, posture symmetry, gaze direction, and temporal behavior patterns. These features serve as inputs to machine learning and deep learning models, enabling objective and quantifiable analysis of behavioral characteristics..
Machine Learning Frameworks	Machine learning frameworks are widely used in autism screening research to build predictive models based on behavioral, clinical, or motion-related features. Studies commonly apply supervised learning techniques to classify children into ASD and non-ASD groups using extracted features from questionnaires, videos, or sensor data.
Deep Learning Architectures	Deep learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are frequently adopted in ASD research to capture complex spatial and temporal behavioral patterns. These architectures are especially effective when working with unstructured data such as videos and sequential motion signals.
Pose Estimation Technologies	Pose estimation technologies are commonly employed to extract skeletal joint coordinates and movement trajectories from video data. In autism-related studies, these tools enable non-invasive analysis of posture, gestures, and repetitive movements without requiring wearable sensors.
Video-Based Computer Vision Systems	Computer vision systems play a central role in analyzing behavioral cues from video recordings. In existing ASD screening research, such systems are used to detect motion patterns, gaze direction, facial expressions, and interaction behaviors that may indicate developmental differences.
Statistical Analysis Software	Statistical tools are widely used in medical AI research to validate model performance and assess clinical reliability. Researchers commonly rely on statistical analysis to compute sensitivity, specificity, accuracy, and confidence intervals when comparing AI predictions with clinical assessments.

3. Chapter 3
Requirement analysis and specifications

3.1. Introduction

The main objective is to explain the system for early detection of autism through using machine learning techniques. The system will serve as a tool to aid in autism screening (not to diagnose) through utilizing machine learning to analyze video footage of patients.

3.2. Work Progress

3.2.1. Research Deliverables

Table 4: Research Deliverables

Deliverables	Description
Source Code	A source code for the model built will be provided.
Dataset	Consists of x videos collected from local behavioural therapy centers
Meta-Data	Anonymized data, such as age and gender, and diagnosis labels for the subjects.
Documentation File	Code and system documentation.
Research Paper	A structured format that includes an introduction, literature review, methodology, results, discussion, and our conclusion.
Statistical Analysis	Analysis of skeletal point features, accuracy metrics
Data Visualization	Skeletal keypoint visualizations, performance graphs
Progress Report	A report that shows the task distribution across team members and the progress of the project will be provided.
Model	A description of the tools used to build the model, along with a description of the testing approach. CNN, LSTM, Multimodal model, Mediapipe pose estimator, OpenPose, OpenCV

3.2.1.1. Analysis Phase

Table 5: Research Tasks Analysis

No.	Task Name	Description	Dependencies	Duration/days
1. Project Initiation and Planning:				
1.1	Define project goals and objectives.	Articulate what the project aims to achieve with clear criteria for measuring progress that aligns well with the goals desired to be met.	-	-
1.2	Identify stakeholders and establish communication channels.	Conduct a thorough analysis of all potential stakeholders that are interested in or may have an impact on the project.	1.1	-
1.3	Create a detailed project plan.	Thoroughly outlining scope, tasks, timeline, milestones, and requirements. Regularly updating the plan to avoid potential risks.	1.1	-
2. Data Collection:				
2.1	Identify relevant datasets for ASD detection.	Review or study related work that may reference datasets related to ASD detection to get an idea of the data needed. Ensuring the collected data is reliable and accurate.	1.1	-
2.2	Obtain the necessary permissions and approvals.	Acquire the necessary permissions from the university, as well as consent forms from parents to begin data collection.	1.1	-
2.3	Collaborate with local centers to collect data	Give specific instructions and specifications to the specialists, with the goal of recreating similar data in the field	2.2	-
3. Data Preprocessing:				
3.1	Processing data into multimodal data	Choosing the optimal approach for dealing with outliers and missing data.	2.3	-
4. Exploratory Data Analysis (EDA):				
4.1	Perform EDA	Gain insights into the dataset to understand it.	3.1	-

4.2	Visualize relationships and patterns in the data.	Represent the data visually to obtain relationships and patterns easier.	4.1	-
5. Feature Engineering:				
5.1	Select relevant features for predicting ASD detection.	Gain an understanding of the domain knowledge to help select the relevant features.	4.2	-
5.2	Transform and create new features.	Ensure that numerical features are scaled and create new useful features from existing ones.	5.1	-

3.2.1.2. Implementation Phase

Table 6: Research Tasks Implementation

No.	Task Name	Description	Dependencies	Duration/days
1. Model Development and Training:				
1.1	Choose machine learning algorithms.	Select the appropriate machine learning algorithms that give the desired outcome.	5.2	-
1.2	Split the dataset and train models.	Split the dataset to training and testing sets.	1.1	-
2. Model Evaluation and Fine-tuning:				
2.1	Evaluate models using appropriate metrics.	Assess performance of the models to see how well the model works, using relevant evaluation metrics.	1.2	-
2.2	Fine-tune models based on evaluation results.	Ensure that the model meets the specific needs and maximize the model's performance.	2.1	-
3. Implementation:				
3.1	Develop machine learning models.	Implement the suitable chosen machine learning models.	2.2	-
3.2	Implement data preprocessing and feature	Explain how data was cleaned, normalized and prepared for training.	3.1	-

	engineering pipelines.			
3.3	Integrate models into the overall system.	Determine where the ML models will be utilized.	3.2	-
1. Testing:				
4.1	Conduct unit testing for individual components.	Verify that each individual component of the model performs as intended.	3.3	-
4.2	Perform system testing to ensure proper integration.	Verify that the model as a whole performs as intended.	4.1	-
5. Manuscript Writing:				
5.1	Methodology	Detailed description of steps taken while conducting ML algorithms, data preprocessing, feature engineering pipelines and information of the dataset used.	4.2	-
5.2	Results & Discussion	Present outcomes of the study including tables, graphs, charts and discuss a report on model performance using appropriate metrics.	5.1	-
2. Publication and Knowledge Sharing:				
6.1	Submit the manuscript to relevant journals or conferences.	Submit the final project after making sure it follows the submission guidelines to relevant journals or conferences to get feedback.	5.2	-

3.2.2. Risk Assessment

3.2.2.1. Analysis Phase Risks

Table 7: Risk Assessment Analysis

No.	Task	Risk	Probability	Impact	Response
1. Project Initiation and Planning:					
1.1	Define project goals and objectives.	Not providing a complete and clear explanation of the idea.	Low	Incomplete requirement, misunderstanding and delays.	Understand and gain a better insight on our project.

		Not setting a goal for the project.			
1.2	Identify stakeholders and establish communication channels.	Misunderstanding stakeholders due to the miscommunication.	Low	Result in project failure .	Create a variety of communication channels
1.3	Create a detailed project plan.	A weak project plan or a project plan with additional useless steps.	Medium	Result in missed requirements, delays and resource overuse.	Regularly review and update our project plan.
2. Data Collection:					
2.1	Identify relevant datasets for ASD detection	Not being able to acquire enough data	Medium	Not enough data to yield significant results, risking overfitting	Start the research for a dataset and clean it early.
2.2	Obtain necessary permissions and approvals.	Not getting consent forms from parents	High	reducing the sample size of our data.	attempt to collect data from as many sources as possible
2.3	Analyze data to understand patterns and trends.	Poor-quality data.	Medium	Inaccurate analysis and misinterpretati on of patterns.	Implement comprehensive data cleaning and preprocessing steps
3. Data Preprocessing:					
3.1	Normalization and Standardization	If not performed, we might get features with larger scales and different units.	Medium	Effect performance of model and result in biased results.	Apply normalization and standardization techniques.
3.2	Handle missing values, outliers, and duplicates.	Missing values can lead to incomplete data, outliers will significantly skew statistical measures, and duplicate values will lead to overfitting.	High	Missing values and outliers will result in biased or inaccurate predictions. Overfitting will lead to poor performance on unseen data.	Use the right methods to handle those issues like statistical methods, or using advanced imputation techniques.

4. Exploratory Data Analysis (EDA):					
4.1	Perform EDA to gain insights into the dataset.	Not understanding where outliers occur and how variables are related.	Medium	Incomplete or inaccurate data. Incorrect identification and handling of outliers.	Look for unusual patterns or trends by visualizing the distribution of each variable.
4.2	Visualize relationships and patterns in the data	Overplotting.	High	Unable to understand the plot and difficult to see individual data points.	Downsample the data and use different colors/sizes.
5. Feature Engineering:					
5.1	Select relevant features for predicting ASD detection	Incomplete feature Set.	High	Poor predictive performance	Ensure that all relevant features are considered.
5.2	Transform and create new features.	Incorrect application of transformations.	Low	Too many features that will lead to overfitting.	Review the impact of each transformation and new features created.

3.2.2.2. Implementation Phase Risks

Table 8: Risk Assessment Implementation

No.	Task	Risk	Probability	Impact	Response
A. Model Development and Training:					
1.1	Choose machine learning algorithms.	Select an unsuitable algorithm.	Low	Poor model performance.	Select a suitable algorithm that aligns with the requirements of our project.
1.2	Split the dataset and train models.	Biased Dataset Split.	Low	Biased model evaluation.	Use stratified sampling.
B. Model Evaluation and Fine-tuning:					
2.1	Evaluate models using appropriate metrics.	Inappropriate evaluation metric	Low	Inaccurate assessment of model performance.	Choose to evaluate metrics that align with the desired outcomes.

2.2	Fine-tune models based on evaluation results.	Can lead to overfitting in the data.	Medium	Overfitting leads to low accuracy.	Apply fine-tuning.
C. Manuscript Writing:					
3.1	Document the entire process.	Insufficient documentation	Low	Challenges in sharing knowledge, addressing upcoming obstacles, or applying the process.	Make organized and understandable documentation.
3.2	Write methodology, results, and discussion.	Methodology lacks clarity.	Low	Difficulty in understanding the procedures followed or steps taken in the study.	Clearly explain the methodology and the steps taken to get the results.
D. Implementation:					
4.1	Develop machine learning models.	Building overly complex models	Low	Cause interpretation issues and possible overfitting in the training set.	Prevent overfitting and start with simple models.
4.2	Implement data preprocessing and feature engineering pipelines.	Feature scaling issues.	Medium	Result in biased model training, particularly in algorithms that are sensitive to feature scaling.	Apply proper feature scaling methods and insure it in the preprocessing pipeline.
4.3	Integrate models into the overall system.	Model complexity.	Medium	Affect system responsiveness and efficiency.	Simplify models using feature selection and dimensionality reduction .
5. Testing:					
5.1	Conduct unit testing for individual components.	Incomplete or insufficient unit test.	Medium	Undetected bugs and impact the performance.	Make sure to unit test all individual components.
5.2	Perform system testing to ensure proper	Failures in system integration	Medium	Poor performance and system not	System testing to evaluate the system's performance under

	integration.			functioning .	various loads and circumstances
6. Publication and Knowledge Sharing:					
6.1	Submit the manuscript to relevant journals or conferences.	Submitting false data or false information.	Low	Rejection or delay in publication.	Make sure all information and data we used in the research is 100% correct.

3.3. Requirement Analysis and Specifications

3.3.1. Business Requirements identification

Obtaining the business requirements for early detection of autism was a vital stage of the system development life cycle. Due to the medical nature of our project we relied on current diagnostic limitations regarding ASD.

Currently, the diagnosis of autism relies heavily on behavioral observation scales (like ADOS-2). Though this has limitations.

1. Diagnosis often depends on the clinician's personal observation skills, which can vary between doctors.
2. Subtle physical signs are often missed which leads to prolonged time in reaching a diagnosis
3. Parents may not have access to specialized centers for diagnosis due to multiple reasons (long wait time for appointment, financial constraint, etc.)

Our system attempts to address these limitations.

1. The solution must function using standard video input (e.g., smartphone or webcam) without requiring the child to wear sensors, which could cause distress.
2. The platform must be accessible via a web interface to allow evaluators to use it on existing clinic computers without specialized hardware installation.
3. The system is not intended to replace the doctor, it is only a tool to assist in diagnosis.

3.3.1.1. User Requirements

Table 9: User Requirements

Functional Requirements	***Non-Functional Requirements***
The evaluator should be able to login securely.	Login process should be relatively quick
The evaluator should be able to create a new patient profile to add age and gender.	Patient profile creation should be intuitive and quick to complete
The evaluator should be able to upload video files of the patient for diagnosis.	Upload progress should be shown in real time

The evaluator should be able to record videos through a webcam.	There should be clear visual indicators for recording status
The evaluator should be able to view the results report of the diagnosis which contains the probability of ASD.	Results should be presented in clear, non-technical language (no need for AI background)
The evaluator should be able to flag a result as incorrect.	Flagging processes should be intuitive and simple

3.3.1.2. System Requirements

- The input video will have adequate lighting and resolution
- The patient's full body will be visible in the video frame
- The model will assume there is only one target in the video (the patient)

Table 10: System Requirements

Functional Requirements	***Non-Functional Requirements***
The system should display a login screen requiring valid credentials.	The system will not store the video data permanently without encryption.
The system should redirect the user to the "Dashboard" upon successful login.	The system should process a 1-3 minute video and return results within 5 minutes maximum.
The system should provide a "Drag and Drop" zone for video uploads.	The underlying model must achieve a minimum sensitivity/recall of 80% to be clinically useful.
The system should validate the file format (check if it is a valid video file) before processing.	The interface should be clean and simple, requiring no technical knowledge of AI to operate.
The system should extract skeletal keypoints (using MediaPipe/OpenPose) from the uploaded video.	If the video quality is too poor for analysis, the system should reject it with a clear error message.
The system should feed these keypoints into the trained Classifier Model.	
The system should calculate a "Risk Score" (0-100%) based on the model output.	

3.3.2. Stakeholder Requirements

Table 11: Stakeholder Requirements

The Evaluator	• A tool is required to provide objective, mathematical quantification of posture to reduce human subjectivity in diagnosis. • The evaluator needs to process long videos rapidly to reduce the time spent manually watching hours of footage. • The evaluator requires a downloadable summary report to include in the patient's official medical file. • The evaluator requires a simple, web-based interface that does not require coding knowledge or installation of complex software.
The Parent/Guardian	• The parent requires strict assurance that their child's video data will not be leaked, shared publicly, or stored longer than necessary for analysis. • The parent requires a way to easily share videos (e.g., via upload) without needing specialized equipment—standard smartphone footage should be enough. • The parent requires a diagnostic method that does not physically disturb or distress the child (no physical attachment of sensors onto the patient's body).

4. Chapter 4
System Design

4.1. Introduction:

The main objective of this chapter is to present a detailed description of the system design of the proposed AI-assisted autism screening platform. System design focuses on defining the system structure, behavior, data flow, and user interaction in order to translate functional requirements into a clear and organized technical solution.

This chapter illustrates the overall architecture of the system using several design models, including context diagrams, use case diagrams, data flow diagrams, class and sequence diagrams, business process models, user interface design, and data modeling. These models collectively describe how the system operates, how users interact with it, how data is processed, and how results are generated and stored.

The proposed system is a web-based application designed to support rehabilitation specialists and occupational therapists in local clinics by analyzing children's posture and behavioral patterns from video recordings using computer vision techniques. The system provides ASD risk scores, confidence levels, and visual annotations, helping specialists perform early screening in a more objective, efficient, and scalable manner. It is important to note that the system does not provide a medical diagnosis, but rather assists specialists during the initial screening and follow-up process.

4.2. System Data Flow Diagrams

A context diagram provides a high-level overview of the system and defines its boundaries by illustrating how external entities interact with it. The proposed system is represented as a single process that communicates with external users and systems.

4.2.1. Context Diagram - Level 0 Data Flow Diagram (DFD)

At Level 0, the system is treated as a black box named **"ASD Screening System"**. The external entities interacting with the system are:

- Rehabilitation Specialist / Occupational Therapist
- System Administrator
- Parent/Child Data Source

The specialist uploads or records videos, views analysis results, and manages child profiles. The system administrator manages users and maintains system functionality. The system outputs analysis results, reports, and visual annotations.

Figure 5: Context Diagram – Level 0 Data Flow Diagram

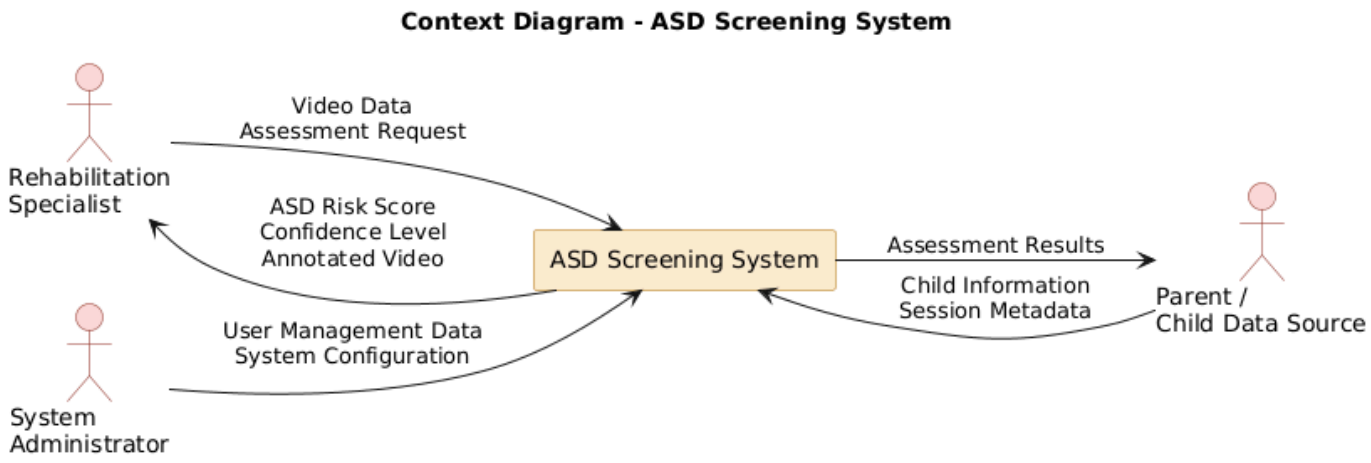


Table 12: Context Diagram – Level 0 Data Flow Diagram

External Entity	***Input to System***	***Output from System***
Rehabilitation Specialist	Video recordings	Risk score, annotations, video, confidence level
System Administrator	User management data System configuration	-
Child Data Source (ex. Parent or guardian)	Session metadata Child information	Assessment Result

4.2.2. Level 1 Data Flow Diagram (DFD)

At Level 1, the internal components of the system are introduced. The system is decomposed into major subsystems:

- User and Profile Management Module
- Video Acquisition and Processing Module
- AI Behavior Analysis Module
- Assessment and Reporting Module
- Database Management System

The rehabilitation specialist interacts with the system through a web-based user interface to manage child profiles, upload or record videos, and review assessment results. The video data is processed by the AI analysis module, and the generated outputs are stored in the database to support historical tracking, specialist annotations, and follow-up assessments.

Figure 6: Level 1 Data Flow Diagram (DFD)

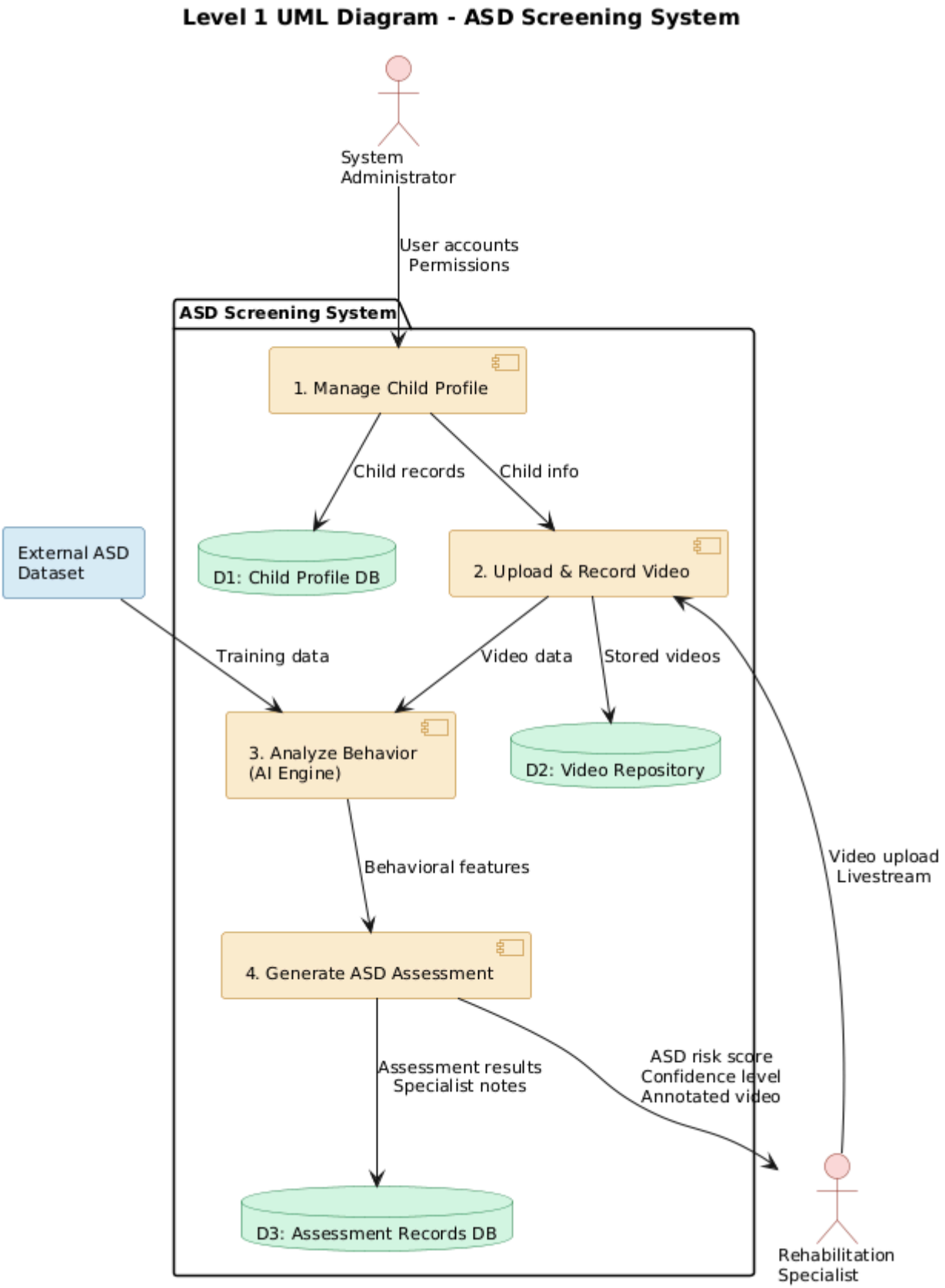


Table 13: Data Flow Diagram – Level 1

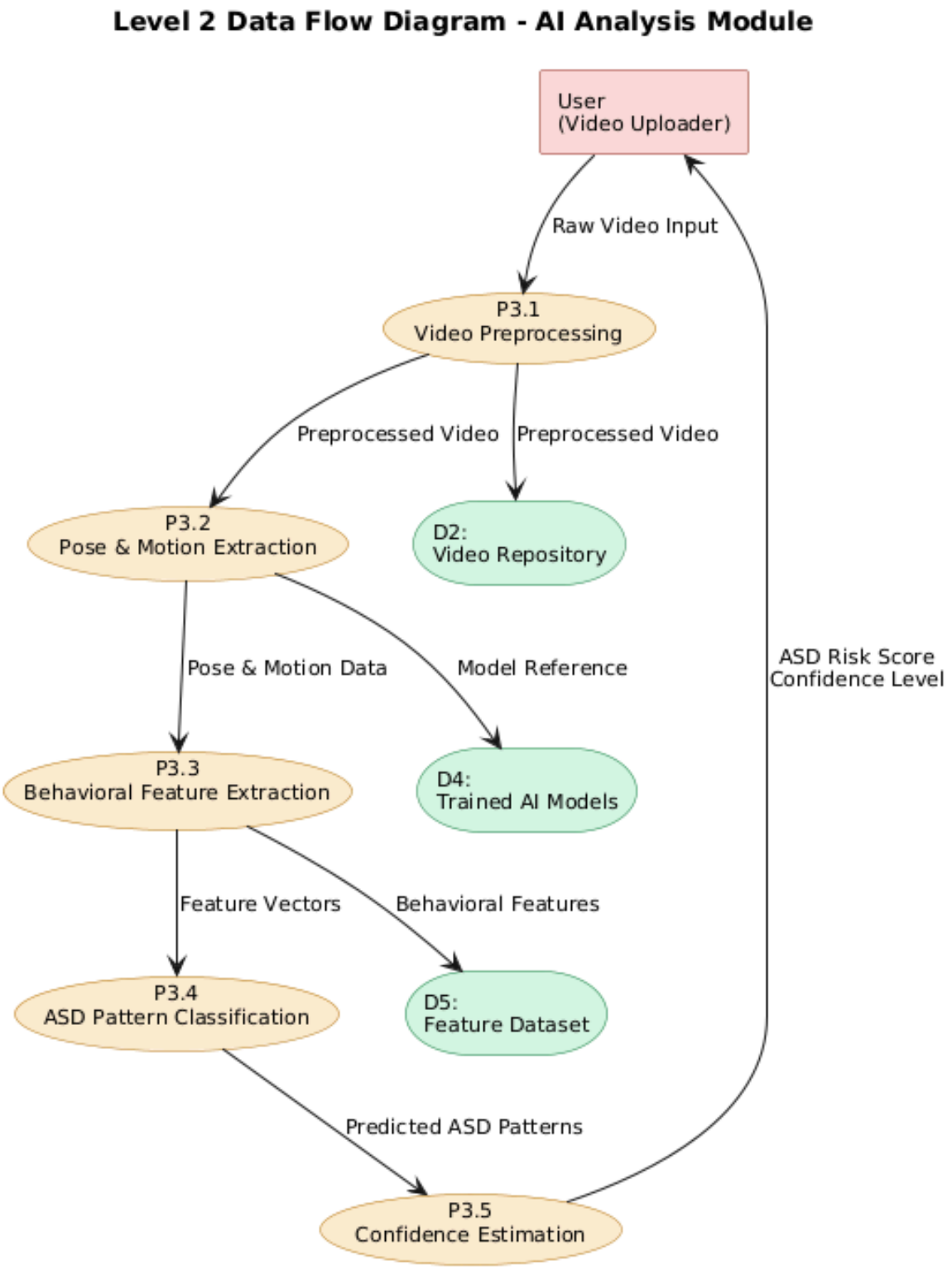
ID	***Component***	***Type***	***Function***
P1	Manage Child Profile	Process	Child profiles, including demographic and developmental information required for ASD screening.
P2	Upload & Record Video	Process	Allows specialists to upload pre-recorded videos or capture live video streams of children during behavioral observation sessions.
P3	Analyze Behavior (AI Engine)	Process	Applies machine learning models trained on ASD datasets to analyze posture and behavioral patterns from uploaded videos.
P4	Generate ASD Assessment	Process	Integrates AI analysis results to generate an ASD risk score, confidence level, and visual behavior annotations.
D1	Child Profile Database	Data store	Stores persistent records of child data.
D2	Video Repository	Data store	Stores uploaded and recorded video files for processing, review, and future comparisons.
D3	Assessment Records Database	Data store	Maintains historical ASD assessment results, specialist notes.
-	External ASD Dataset	-	Dataset created for the purpose of training the AI machine.

4.2.3. Level 2 Data Flow Diagram (DFD)

At Level 2, the internal processes are further detailed. This level demonstrates how video data flows through the system until meaningful results are produced and presented to the specialist. A Level-2 diagram was created only for the AI Behavior Analysis Module (P3), since it is the system's core AI component. Focusing on P3 highlights the complex feature extraction and model-based decision-making central to data science and AI explainability, while other modules are more straightforward and do not require detailed decomposition.

4.2.3.1. Level 2 Context Diagram for AI Behavior Analysis Module

Figure 7: Level 2: In-depth DFD for Process 3

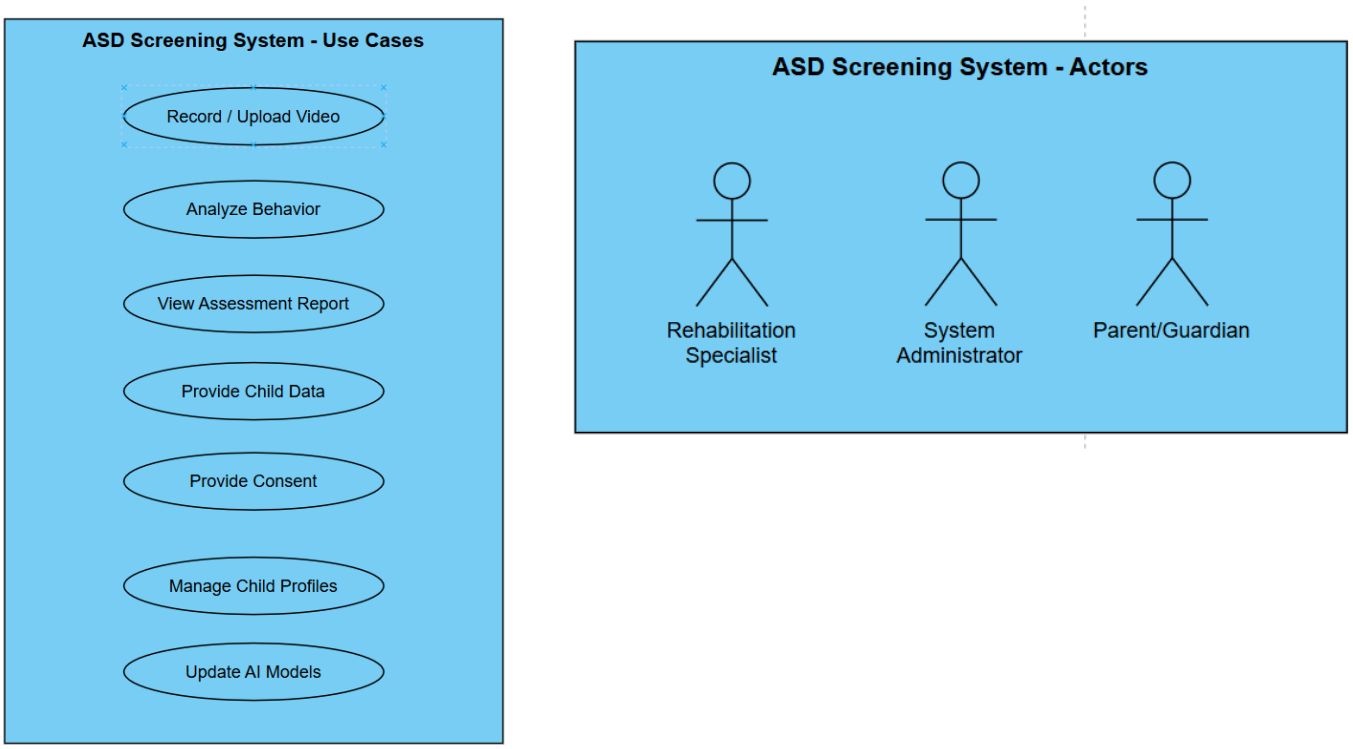


4.3. *Use Case Diagram*

This chapter presents the system architecture using various design models to describe system functionality, user interactions, data processing, and result generation.

A use case diagram, defined by the Unified Modeling Language (UML), illustrates the functional behavior of the system from the users' perspective by showing system actors and the use cases they perform. This section presents a consolidated use case diagram that captures all actors interacting with the system and their associated functionalities.

Figure 8: ASD Screening System - Use Cases & Actors



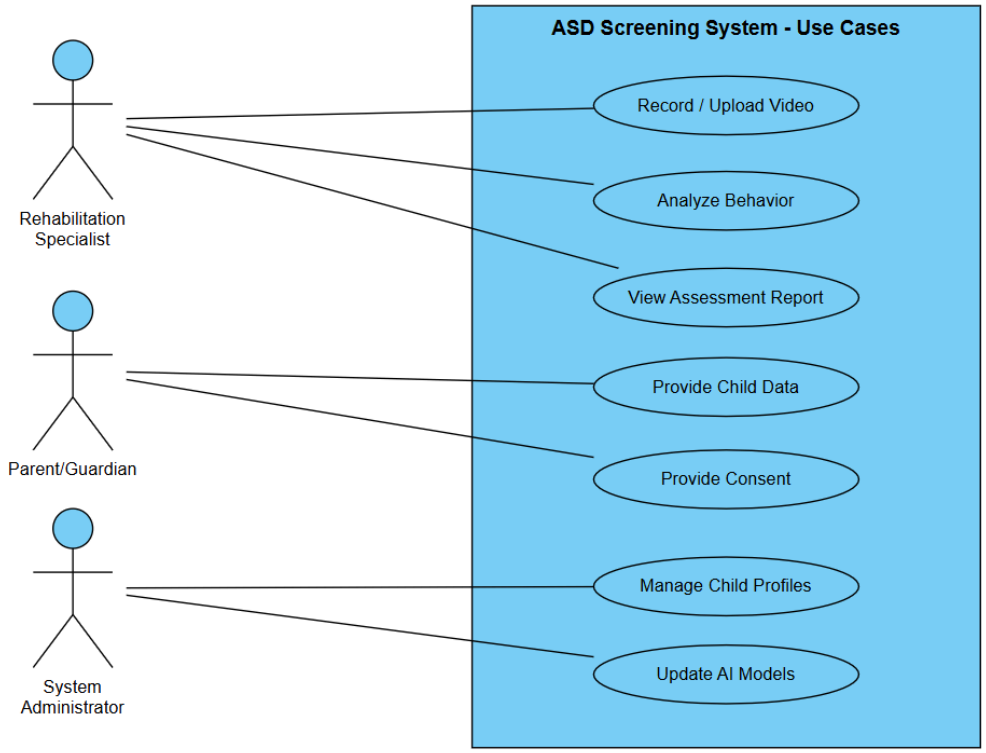
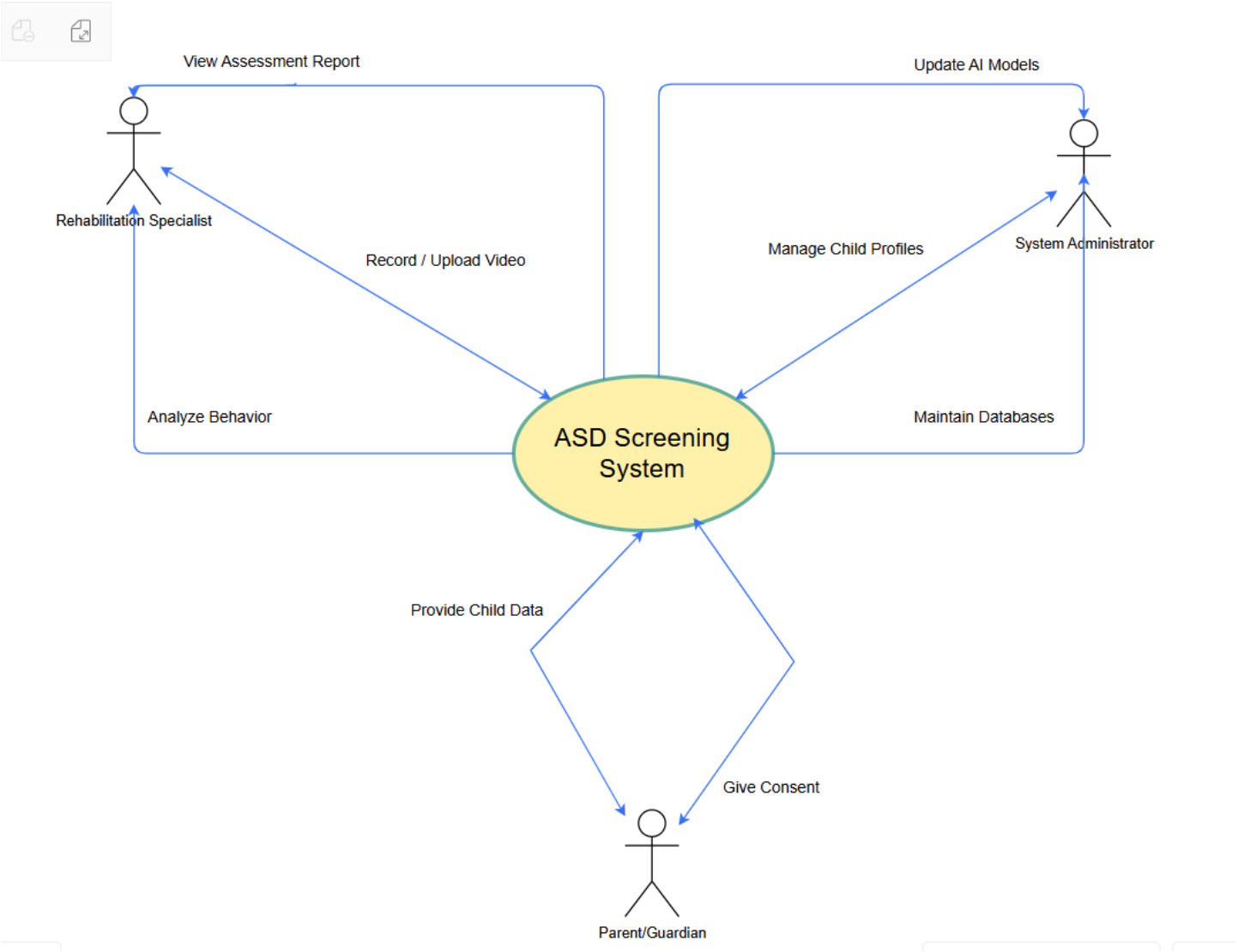


Figure 9: Use Case Diagram of the ASD Screening System



This system will include 8 different use cases shown in the Use Case Diagrams above. They are:

Table 14.1: Use Case 1 - Record / Upload Video

Actor	***System Response***
Rehabilitation Specialist	Provides an interface to record or upload a child's video during a therapy session, validates the video format and quality, and securely stores the video in the video repository.

Table 14.2: Use Case 2 - Analyze Behavior

Actor	***System***
Rehabilitation Specialist	Processes the uploaded video using AI-based computer vision and machine learning techniques to extract behavioral features, classify ASD-related patterns, and generate confidence scores.

Table 14.3: Use Case 3 - View Assessment Report

Actor	***System***
Rehabilitation Specialist	Displays the generated assessment report, including ASD risk score, visual annotations, timestamps, and summary findings for clinical review.

Table 14.4: Use Case 4 - Provide Child Data

Actor	***System***
Parent/Guardian	Receives and records child-related information and behavioral context required for assessment and securely links the data to the child's profile.

Table 14.5: Use Case 5 - Provide Consent for Data Usage

Actor	***System***
Parent/Guardian	Records informed consent for video recording, data processing, and AI-based analysis in compliance with ethical and privacy requirements.

Table 14.6: Use Case 6 - Manage Child Profiles

Actor	***System***
System Administrator	Creates, updates, or removes child profiles and ensures the integrity and consistency of stored data.

Table 14.7: Use Case 7 - Update AI Models

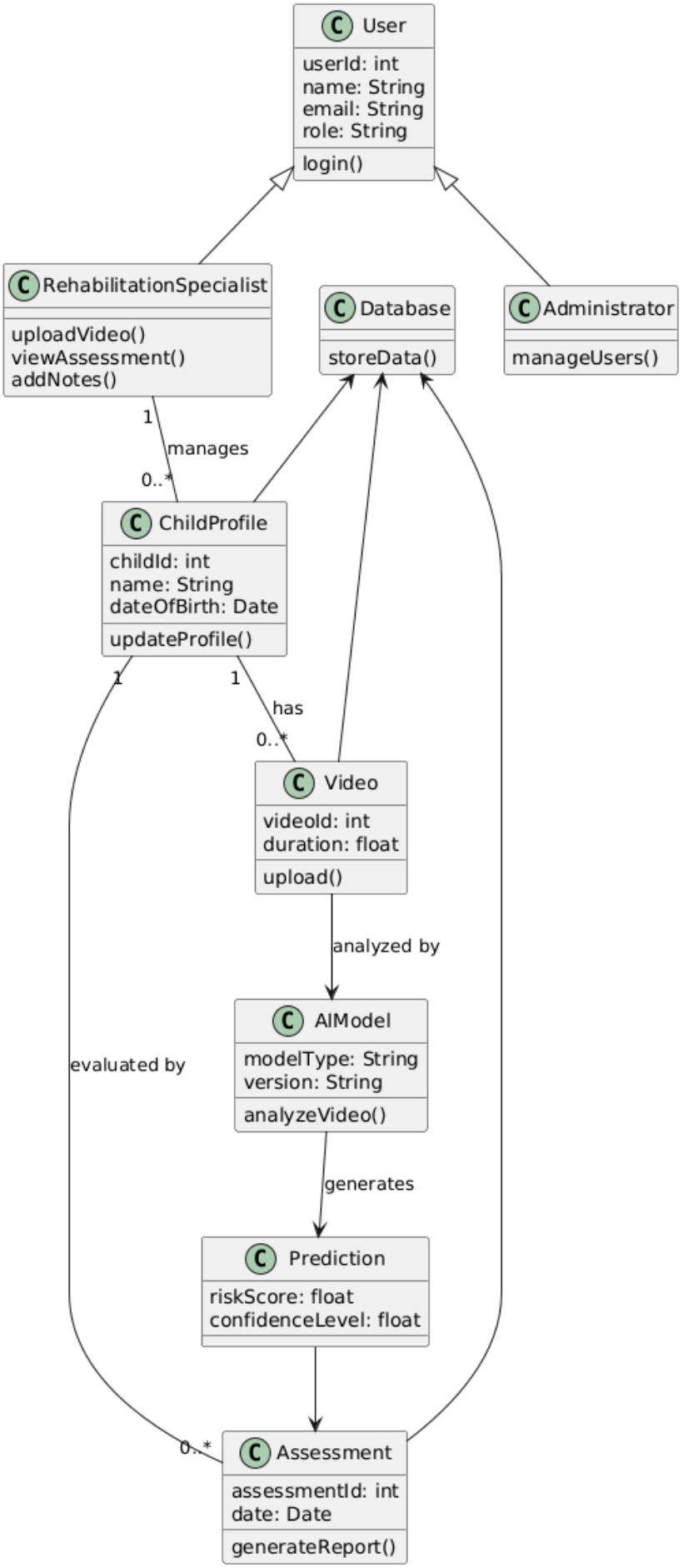
Actor	***System***
System Administrator	Uploads, retrains, or updates AI models used for behavior analysis and logs model changes for traceability.

Table 14.8: Use Case 8 - Maintain Databases

Actor	***System***
System Administrator	Allows the administrator to add, update, or delete records in the child profiles, video repository, and assessment database. It ensures data integrity, logs all changes, and confirms successful operations.

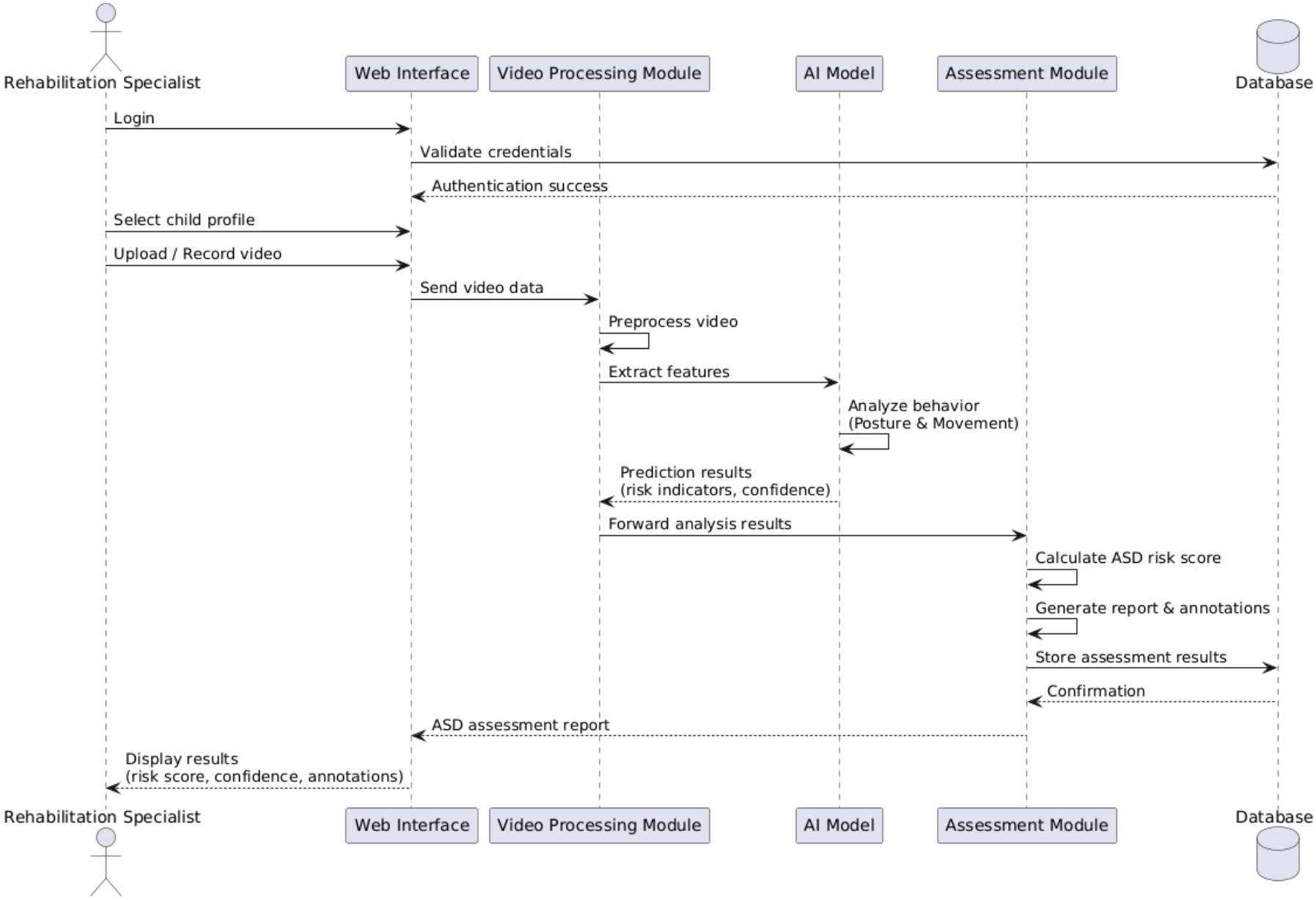
4.4. Class Diagram

Figure 10: Class Diagram for ASD Screening System



4.5. *Sequence Diagram*

Figure 11: Sequence Diagram for ASD Screening System

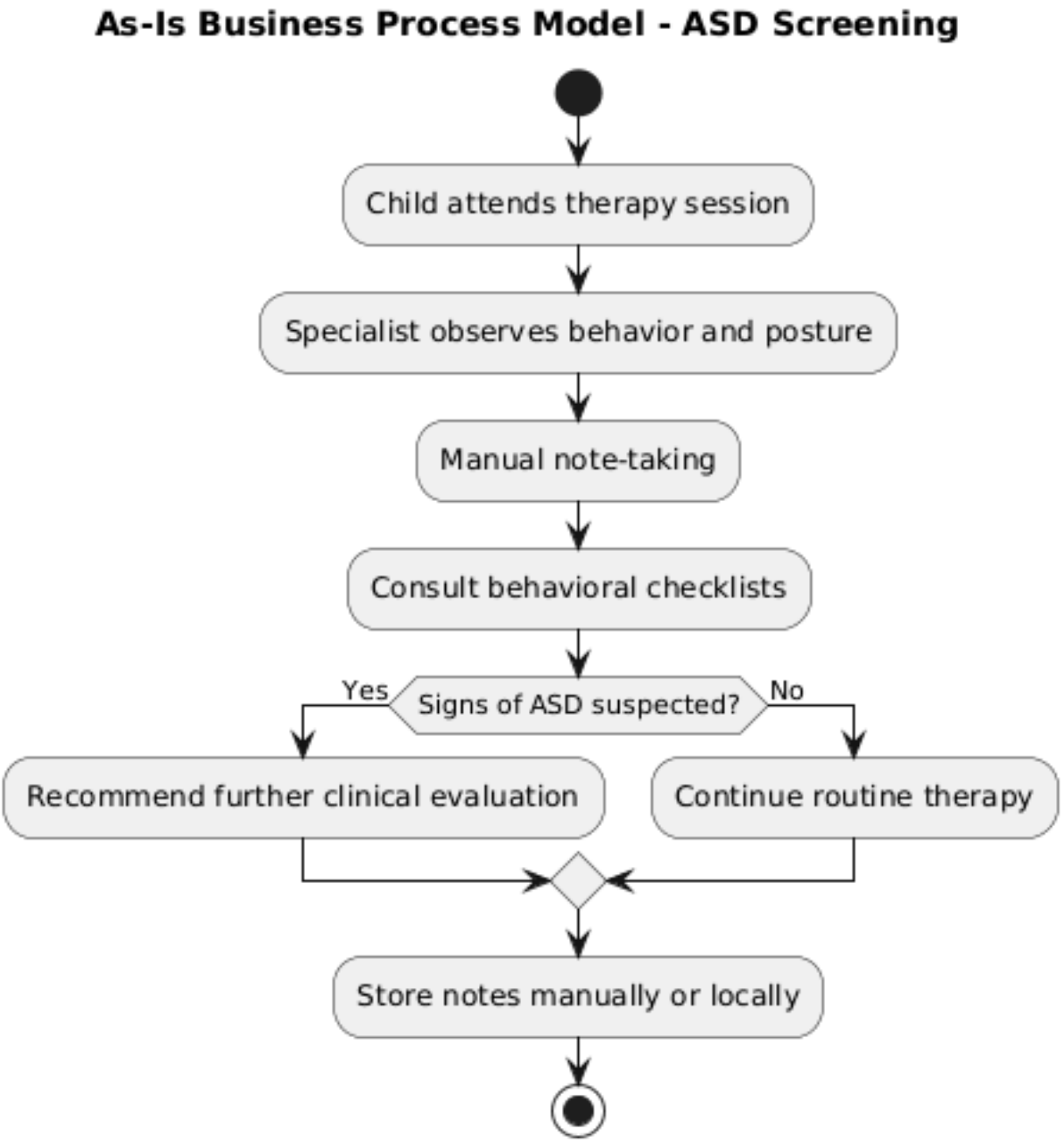


4.6. Business Process Model

4.6.1. As-Is Business Process Model

In the current (As-Is) process, ASD screening relies heavily on manual observation and specialist expertise. The rehabilitation specialist observes the child during therapy sessions and relies on personal experience, behavioral checklists, and parental feedback to identify potential ASD indicators. The assessment process is time-consuming, subjective, and limited by specialist availability. Records are often stored manually or in isolated digital formats, making follow-up comparisons difficult.

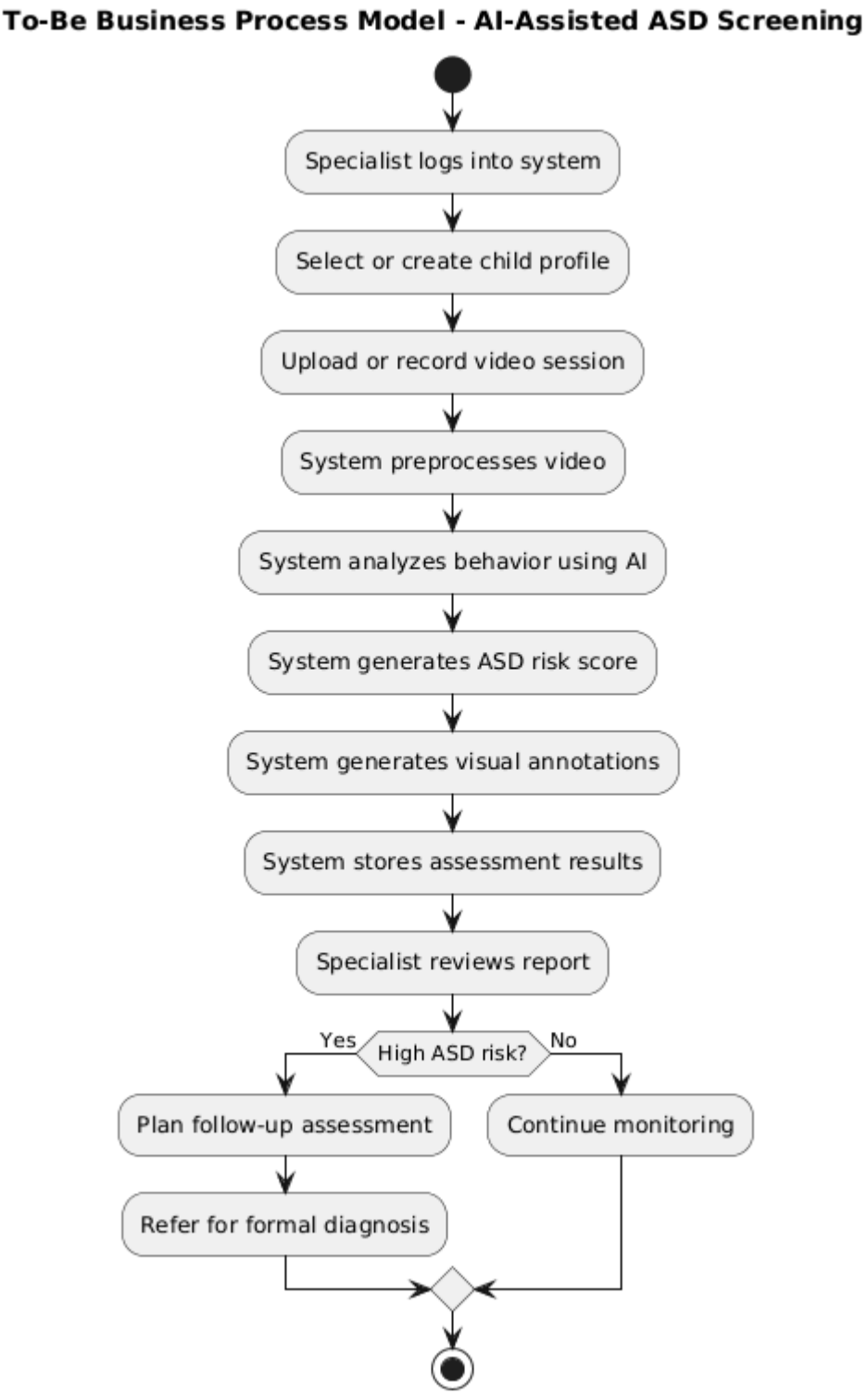
Figure 12: As-Is Business Process Model for ASD Screening



4.6.2. To-Be Business Process Model

In the proposed (To-Be) model, the ASD screening process is enhanced through the integration of an AI-based web system. The rehabilitation specialist records or uploads video sessions of the child, which are automatically analyzed using computer vision and machine learning techniques. The system generates an ASD risk score, visual behavioral annotations, and a confidence level to support clinical decision-making. All assessments are securely stored, enabling follow-up evaluations and longitudinal comparison, thereby improving efficiency, objectivity, and scalability.

Figure 13: To-Be Business Process Model for ASD Screening



4.6.3. Comparision

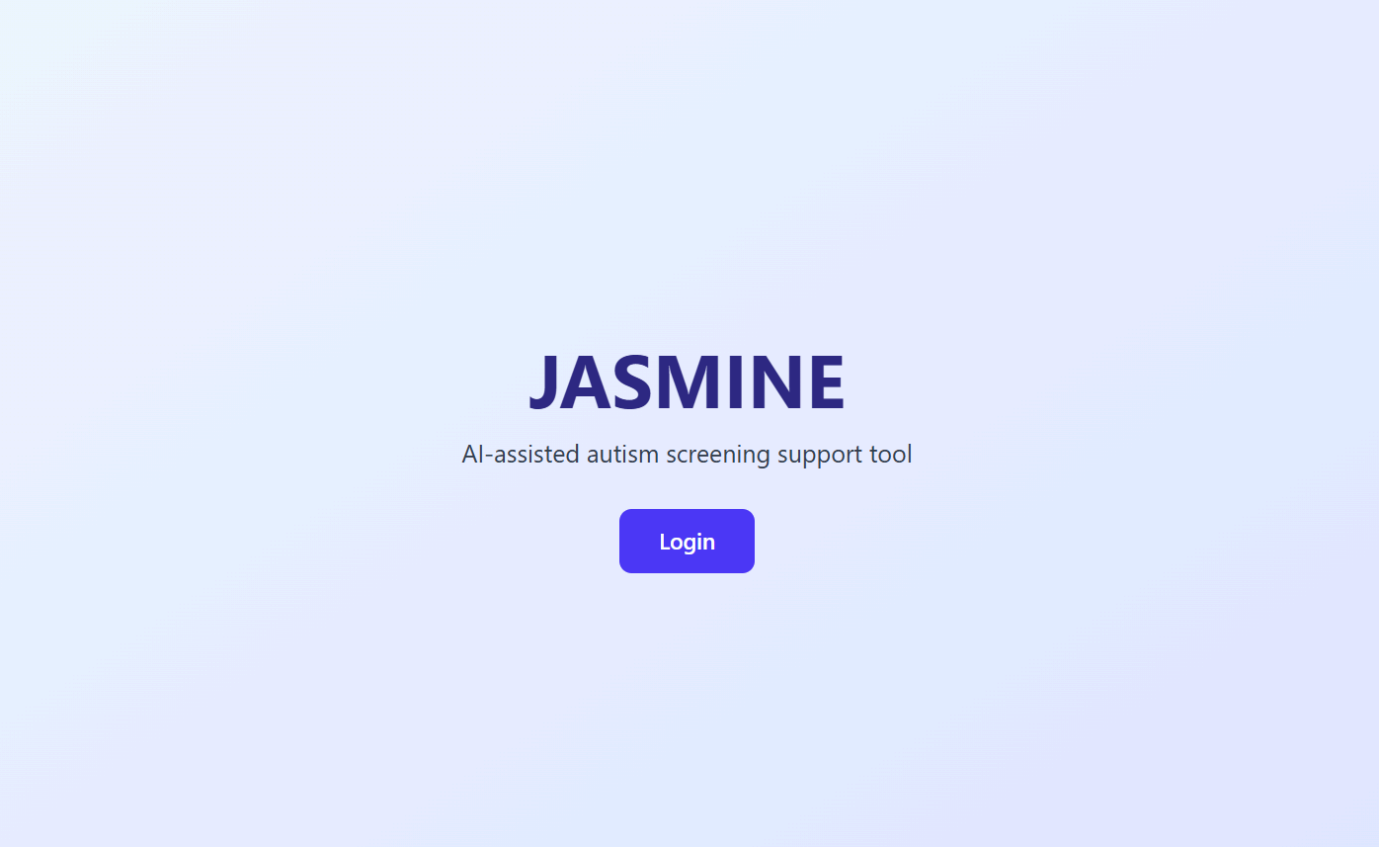
Table 15: As-Is vs To-Be Model Comparison

Aspect	***As-Is Model***	***To-Be Model***
Screening Method	Manual observation	AI-assisted video analysis
Objectivity	Subjective	Data-driven
Time Efficiency	Time-consuming	Faster assessment
Scalability	Limited by specialists	Supports larger number of children
Data Storage	Manual/fragmented	Centralized database (more data = higher cost)
Follow-up	Difficult	Longitudinal comparison supported

4.7. User Interface Design

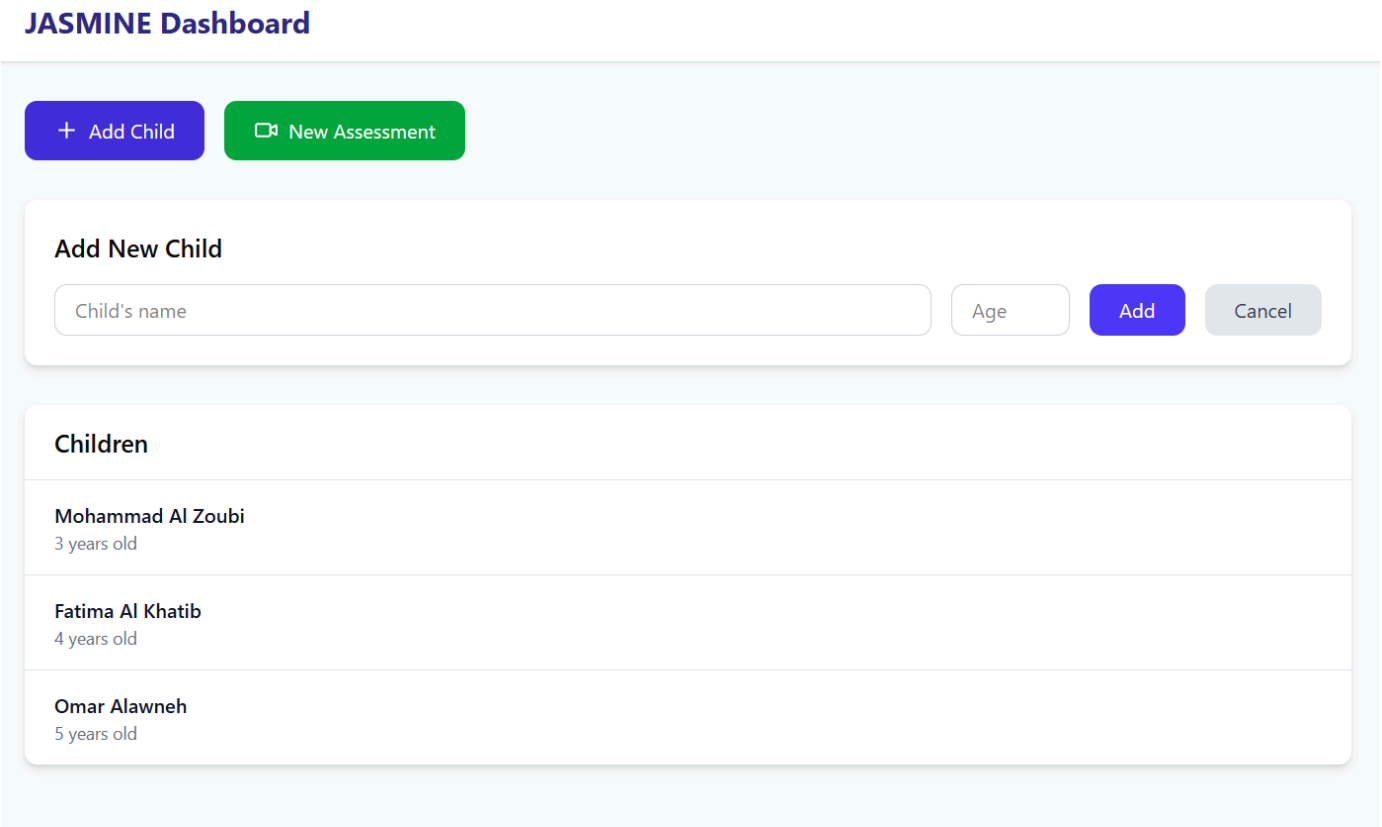
The user interface presented in this section represents a conceptual design intended to illustrate the main system interactions. As the JASMINE system has not yet been implemented, the interface focuses on simplicity and core functionality rather than visual design or commercial usability.

1. Page 1: Home Page

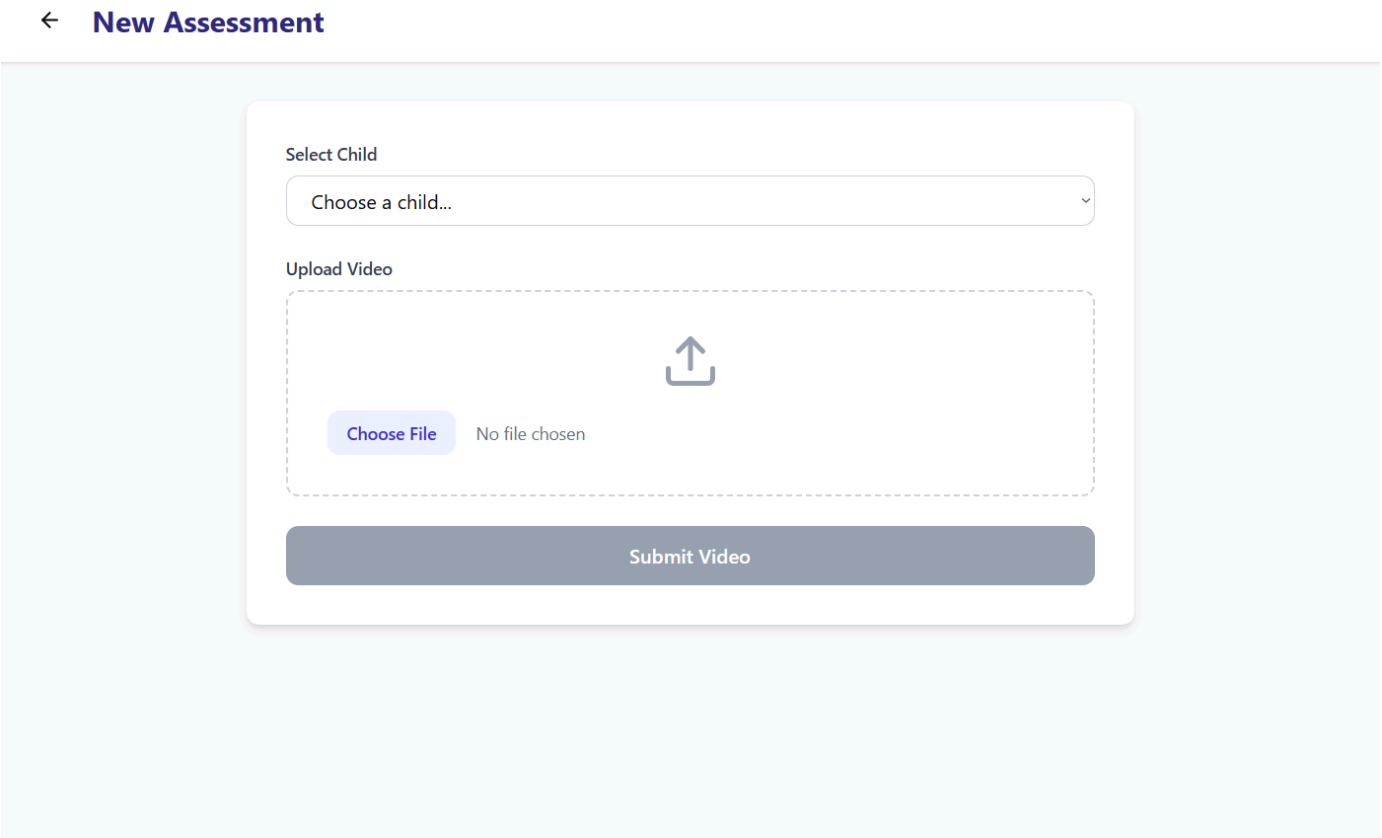


2. Page 2: Login Page

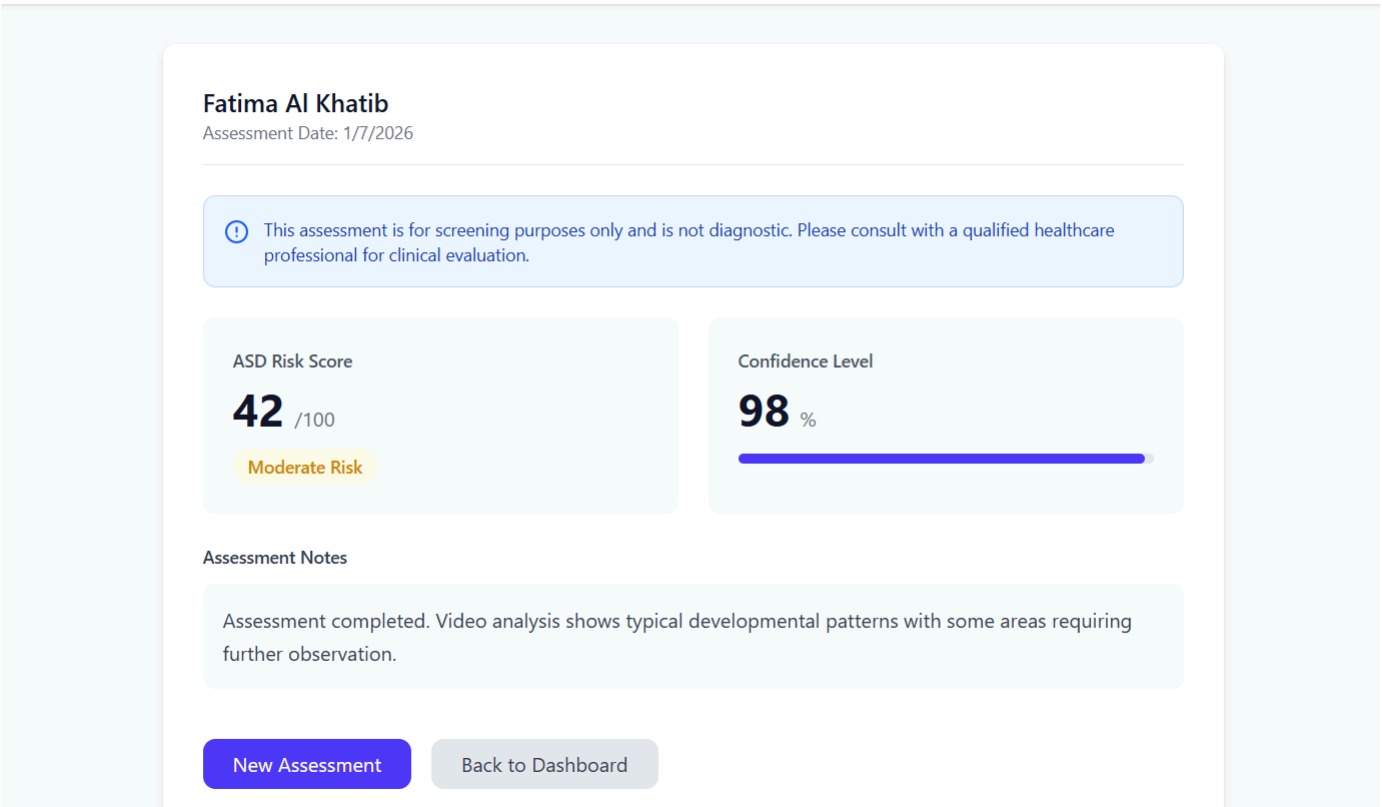
3. Page 3: Dashboard Page



4. Page 4: Video Upload Page (Assessment Request)

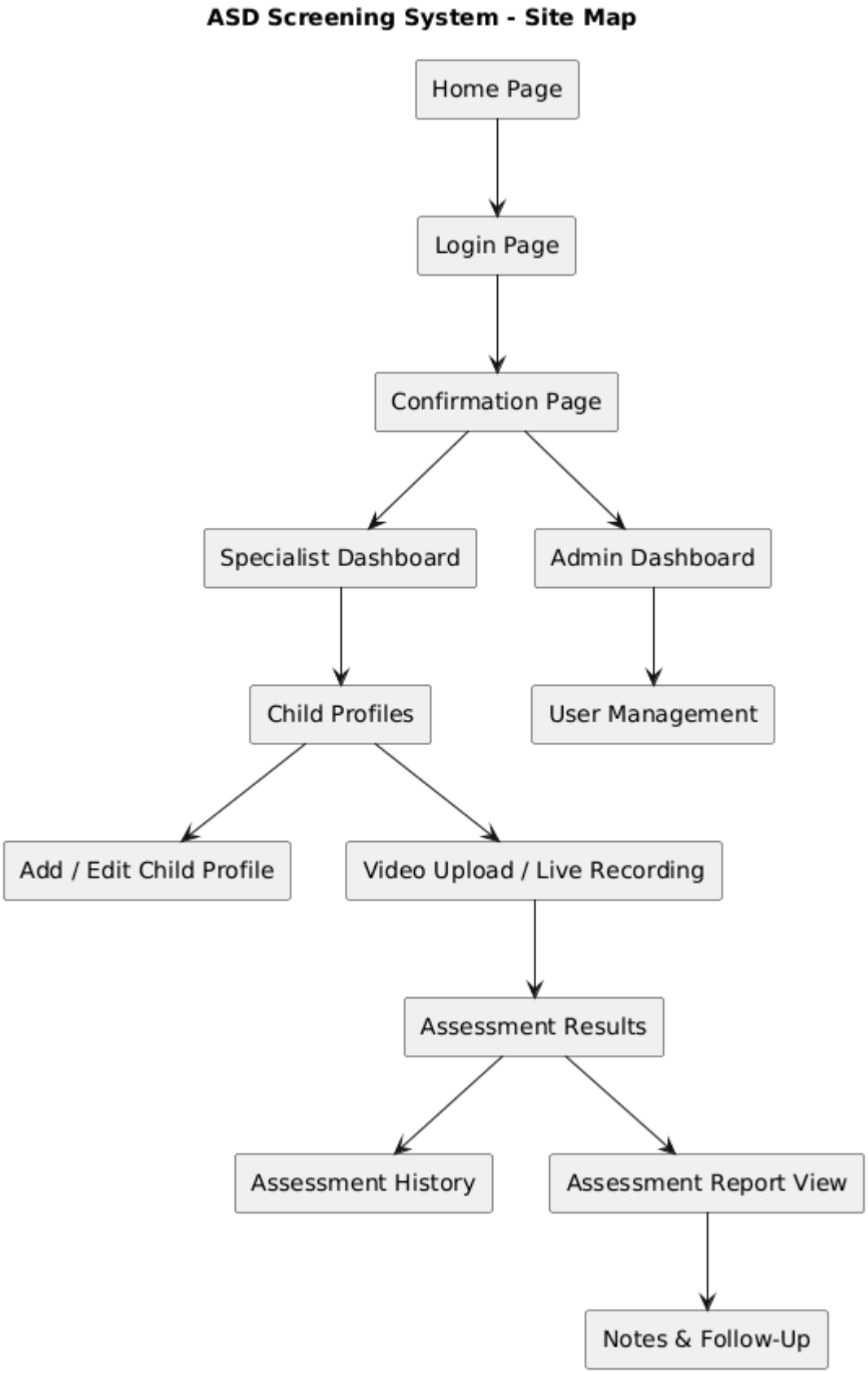


5. Page 5: Assessment Results Page



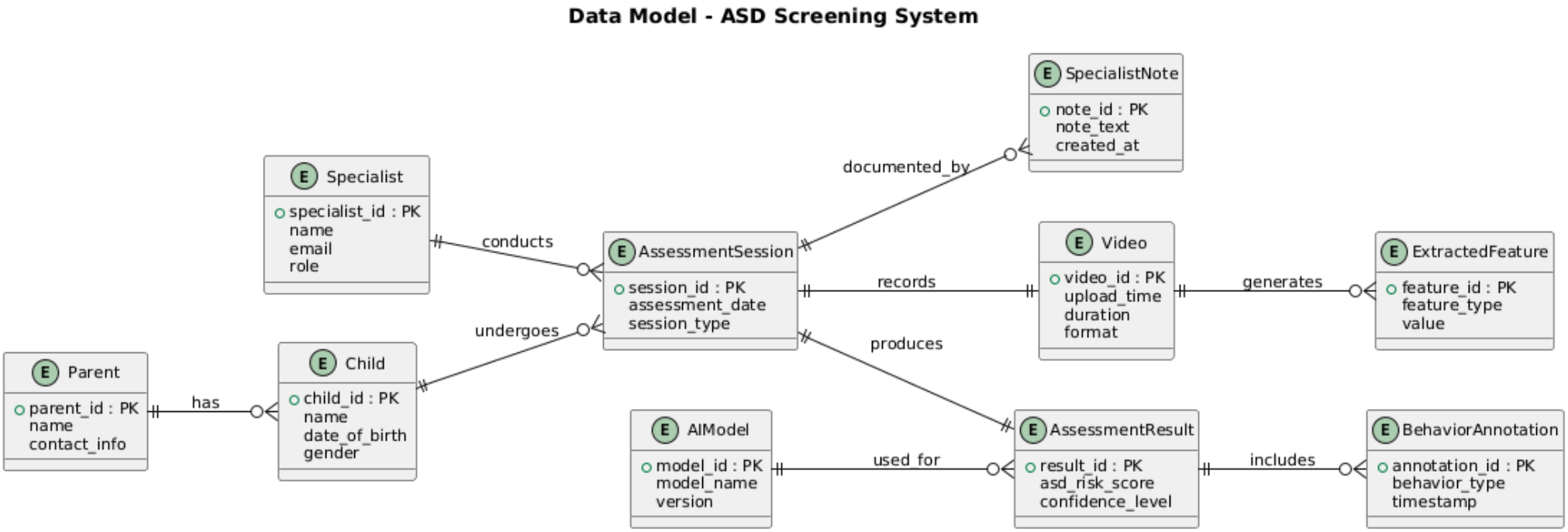
4.7.1. *Site Map*

Figure 14: ASD Screening System Site Map



4.8. Data Map

Figure 15: ASD Screening System Data Map



5. Chapter 5
Conclusion

5.1. Conclusion

In conclusion, the JASMINE system aims to improve the complex, time-consuming process that comes with autism screening, as well as encourage the Jordanian population to screen their children early when symptoms are first noticed. As with many other diseases and disorders, early intervention is vital in reducing the effects autism can have later on a child's development and quality of life. Our goal is to assist rehabilitation specialists, therapists, parents, and children in creating a smoother, technology-assisted screening process.

In this document, we've presented the main steps that will lead to the creation of the system, starting from extensive research and strengthening our domain knowledge, to specifying the requirements needed to accomplish the design.

5.2. Difficulty Faced

During the development and design of the JASMINE system, several challenges were encountered across different stages of the project. These difficulties are summarized as follows:

- ***Limited Availability of Local Data:*** One of the main challenges we faced initially when coming up with the idea was the lack of publicly available data out there. As you can probably guess, there aren't datasets of children with ASD readily available on the Internet, let alone *Jordanian* children. Thus, obtaining consent-compliant recordings from rehabilitation centers required significant coordination with our department and local clinics.
- ***Ethical and Privacy Concerns:*** Adding to our previous point, the handling of child data raised ethical, legal, and privacy concerns. Ensuring data confidentiality, secure storage, and compliance with ethical research standards added complexity to the system design and data management processes. This is why it was important to use feature extraction in our preprocessing pipeline, as it would ensure no raw video is stored in our database.
- ***Limited Domain Expertise:*** Though we might have had some general knowledge prior to starting the project, autism screening is a highly specialized medical domain. Therefore, Gaining sufficient understanding of clinical screening practices, behavioral indicators, and assessment criteria required extensive research.

5.3. Future Works

Future work will primarily involve the implementation and deployment of the proposed system based on the presented architecture and models. This includes developing the web-based platform, integrating video upload and storage mechanisms, and implementing the artificial intelligence models for posture, motion, and behavioral analysis. Further work will also involve collaboration with local rehabilitation centers to collect ethically approved and anonymized video data for training, testing, and validating the AI models.

Through this approach, JASMINE aspires to contribute positively to early autism screening efforts within the Jordanian healthcare and rehabilitation ecosystem.

6. Chapter 6

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